

JESS Thermodynamic Database v8.9

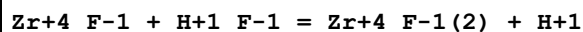
Zirconium

24-Jan-23

Reaction No. 2739							
Zr+4 + H+1_F-1 = Zr+4_F-1 + H+1							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:7.0 (2SF)	1	EOR	
2	25	3	(H,Na)ClO4	lgK:4.89 (3SF)	0	MDS	140
3	25	3	NaClO4	lgK:4.09 (3SF)	0	MDS	437
4	25	4	(H,Na)ClO4	lgK:5.970 (0.002MD)	3	EOD	20829
5	25	4	NaClO4	lgK:5.50 (3SF)	0	MIS	5077
6	25	4	NaClO4	lgK:5.97 (3SF)	0	MIS	5398
7	20	4	HClO4	lgR:5.9 (0.07SD)	0	CMN	437
8	20	4	HClO4	lgR:5.51 (3SF)	0	MCE	437
9	20	4	HClO4	lgR:5.88 (3SF)	0	MDS	437
10	20	4	HClO4	lgR:5.52 (3SF)	0	MRX	437
11	20	4	HClO4	lgR:5.8 (0.3MD)	0	MIS	5077
12	20	4	HClO4	lgR:5.9 (0.2MD)	0	MIS	5077
13	20	4	HClO4	lgR:5.9 (0.4MD)	0	MSE	13458
14	20	4	HClO4	lgR:5.89 (0.02MD)	3	EOD	20829
15	20	4	HClO4	lgR:5.96 (3SF)	0	MCE	22616
16	25	0.96	HClO4	lgR:5.79 (3SF)	0	MCE	7998
17	25	1	HClO4	lgR:5.3 (0.05SD)	0	MIX	5398
18	25	1	HClO4	lgR:5.40 (0.01MD)	3	EOD	20829
19	25	1	HNO3	lgR:5.4 (0.05SD)	0	MIX	5398
20	25	2	HClO4	lgR:5.81 (3SF)	0	MIX	437
21	25	2	HClO4	lgR:5.80 (3SF)	0	MDS	7998
22	25	2	HClO4	lgR:5.8 (0.4MD)	3	EOD	20829
23	25	2	HClO4	lgR:5.81 (0.02MD)	3	EOD	20829
24	25	2	HNO3	lgR:5.97 (3SF)	0	MCE	5398
25	25	2.88	HClO4	lgR:5.83 (3SF)	0	MCE	7033
26	25	3	HClO4	lgR:4.2 (2SF)	0	ENS	437
27	27	2	HClO4	lgR:4.62 (3SF)	0	MDS	437
28	25	4	HClO4	dG:-33.932 (1.SD) kJ	0	MMX	4923

29	25	4	HClO ₄	dH: -17.52 (1.SD) kJ	0	MMX	4923
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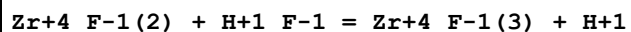
Reaction No. 2740



No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK: 5.3 (2SF)	1	EOR	
2	25	3	(H,Na)ClO ₄	lgK: 3.67 (3SF)	0	MDS	140
3	25	3	NaClO ₄	lgK: 3.67 (3SF)	0	MDS	437
4	25	4	(H,Na)ClO ₄	lgK: 4.37 (0.03MD)	3	EOD	20829
5	25	4	NaClO ₄	lgK: 4.4 (2SF)	0	MIS	5398
6	20	4	HClO ₄	lgR: 4.4 (0.07SD)	0	CMN	437
7	20	4	HClO ₄	lgR: 3.7 (2SF)	0	MCE	437
8	20	4	HClO ₄	lgR: 4.54 (3SF)	0	MCE	437
9	20	4	HClO ₄	lgR: 4.36 (3SF)	0	MDS	437
10	20	4	HClO ₄	lgR: 4.04 (3SF)	0	MRX	437
11	20	4	HClO ₄	lgR: 4.3 (0.1MD)	0	MIS	5077
12	20	4	HClO ₄	lgR: 4.4 (0.4MD)	0	MIS	5077
13	20	4	HClO ₄	lgR: 4.41 (3SF)	0	MCE	7033
14	20	4	HClO ₄	lgR: 4.36 (0.3MD)	0	MSE	13458
15	20	4	HClO ₄	lgR: 3.8 (0.3MD)	0	EOD	20289
16	20	4	HClO ₄	lgR: 4.36 (0.06MD)	3	EOD	20829
17	20	4	HClO ₄	lgR: 4.41 (0.16MD)	3	EOD	20829
18	20	4	HClO ₄	lgR: 4.5 (2SF)	0	MCE	22616
19	25	0.96	HClO ₄	lgR: 4.54 (3SF)	0	MCE	7033
20	25	1	HClO ₄	lgR: 3.8 (0.05SD)	0	MIX	5398
21	25	1	HNO ₃	lgR: 4.2 (0.05SD)	0	MIX	5398
22	25	2	HClO ₄	lgR: 4.3 (0.05SD)	0	MDS	140
23	25	2	HClO ₄	lgR: 3.57 (3SF)	0	MCE	437
24	25	2	HClO ₄	lgR: 4.29 (3SF)	0	MDS	7033
25	25	2	HClO ₄	lgR: 4.32 (3SF)	0	MDS	7998
26	25	2	HClO ₄	lgR: 4.22 (0.11MD)	3	EOD	20829
27	25	2	HNO ₃	lgR: 3.59 (3SF)	0	MCE	437
28	25	2.88	HClO ₄	lgR: 4.51 (3SF)	0	MCE	7033
29	25	3	HClO ₄	lgR: 3.63 (3SF)	0	ENS	437
30	25	4	HClO ₄	lgR: 4.4 (2SF)	0	MIS	5398
31	25	4	HClO ₄	dG: -24.665 (1.SD) kJ	0	MMX	4923

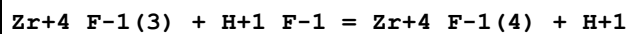
32	25	4	HClO ₄	dH: -16.83 (4SF) kJ	0	MMX	4923
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Reaction No. 2741



No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK: 3.0 (2SF)	1	ELC	
2	25	0.5	NH ₄ ClO ₄	lgK: 2.7 (0.1MD)	0	EOD	20829
3	25	0.5	NaClO ₄	lgK: 2.7 (0.05SD)	0	MRX	140
4	25	3	(H,Na)ClO ₄	lgK: 2.97 (3SF)	0	MDS	140
5	25	3	NaClO ₄	lgK: 2.97 (3SF)	0	MDS	437
6	20	4	HClO ₄	lgR: 3.0 (0.07SD)	0	MDS	437
7	20	4	HClO ₄	lgR: 3.04 (3SF)	0	MRX	437
8	20	4	HClO ₄	lgR: 3.00 (3SF)	0	MCE	7033
9	20	4	HClO ₄	lgR: 3.00 (3SF)	0	MCE	13458
10	20	4	HClO ₄	lgR: 3.0 (0.2MD)	0	MSE	13458
11	20	4	HClO ₄	lgR: 3.00 (0.09MD)	0	EOD	20829
12	20	4	HClO ₄	lgR: 3.14 (0.05MD)	0	EOD	20829
13	25	0.5	HClO ₄	lgR: 2.70 (3SF)	0	MCE	7033
14	25	0.96	HClO ₄	lgR: 2.96 (3SF)	0	MCE	7033
15	25	2	HClO ₄	lgR: 2.8 (0.05SD)	0	MDS	5398
16	25	2	HClO ₄	lgR: 2.88 (3SF)	0	MDS	7033
17	25	2	HClO ₄	lgR: 2.83 (3SF)	0	MDS	7998
18	25	2	HClO ₄	lgR: 3.1 (0.7MD)	0	EOD	20829
19	25	2.88	HClO ₄	lgR: 3.10 (3SF)	0	MCE	7033
20	25	3	HClO ₄	lgR: 2.21 (3SF)	0	ENS	437
21	25	4	HClO ₄	dG: -16.898 (1.SD) kJ	0	MMX	4923
22	25	4	HClO ₄	dH: -11.26 (1.SD) kJ	0	MMX	4923

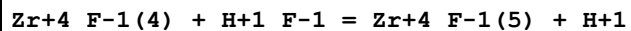
Reaction No. 2742



No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK: 2.4 (2SF)	1	ELC	
2	25	0.5	NH ₄ ClO ₄	lgK: 1.73 (0.07MD)	0	EOD	20829
3	25	0.5	NaClO ₄	lgK: 1.8 (0.05SD)	0	MRX	140
4	25	1	(H,Na)ClO ₄	lgK: 2.80 (0.07MD)	0	EOD	20829
5	25	3	(H,Na)ClO ₄	lgK: 2.77 (3SF)	0	MDS	140

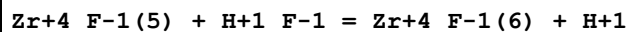
6	25	3	NaClO ₄	lgK:2.77 (3SF)	0	MDS	437
7	20	4	HClO ₄	lgR:2.3 (0.07SD)	0	MDS	437
8	20	4	HClO ₄	lgR:2.20 (3SF)	0	MRX	437
9	20	4	HClO ₄	lgR:2.28 (3SF)	0	MCE	7033
10	20	4	HClO ₄	lgR:2.28 (3SF)	0	MCE	13458
11	20	4	HClO ₄	lgR:2.2 (2SF)	0	MSE	13458
12	20	4	HClO ₄	lgR:2.11 (0.05MD)	0	EOD	20829
13	20	4	HClO ₄	lgR:2.28 (0.13MD)	0	EOD	20829
14	25	0.5	HClO ₄	lgR:1.83 (3SF)	0	MCE	7033
15	25	0.96	HClO ₄	lgR:2.44 (3SF)	0	MCE	7033
16	25	2.88	HClO ₄	lgR:2.36 (3SF)	0	MCE	7033
17	25	3	HClO ₄	lgR:1.94 (3SF)	0	ENS	437
18	25	4	HClO ₄	dG:-12.61 (1.SD) kJ	0	MMX	4923
19	25	4	HClO ₄	dH:-22.1 (1.SD) kJ	0	MMX	4923

Reaction No. 2743



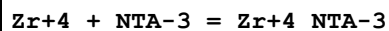
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:1.4 (2SF)	1	ELC	
2	25	0.5	NH ₄ ClO ₄	lgK:1.51 (0.07MD)	0	EOD	20829
3	25	0.5	NaClO ₄	lgK:1.5 (0.05SD)	0	MRX	140
4	25	1	(H,Na)ClO ₄	lgK:1.90 (0.07MD)	0	EOD	20829
5	25	3	(H,Na)ClO ₄	lgK:1.55 (3SF)	0	MDS	140
6	25	3	NaClO ₄	lgK:1.55 (3SF)	0	MDS	437
7	20	4	HClO ₄	lgR:1.5 (0.07SD)	0	MDS	437
8	20	4	HClO ₄	lgR:1.7 (2SF)	0	MRX	437
9	20	4	HClO ₄	lgR:1.53 (0.07MD)	0	MDS	7033
10	20	4	HClO ₄	lgR:1.8 (2SF)	0	MSE	13458
11	20	4	HClO ₄	lgR:1.53 (0.15MD)	0	EOD	20829
12	20	4	HClO ₄	lgR:1.83 (0.04MD)	0	EOD	20829
13	25	0.5	HClO ₄	lgR:1.51 (3SF)	0	MCE	7033
14	25	0.96	HClO ₄	lgR:1.40 (3SF)	0	MCE	7033
15	25	2.88	HClO ₄	lgR:1.56 (3SF)	0	MCE	7033

Reaction No. 2744



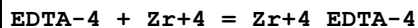
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0.5	NH ₄ ClO ₄	lgK:0.85 (0.04MD)	3	EOD	20829
2	25	0.5	NaClO ₄	lgK:0.9 (0.05SD)	0	MRX	140
3	25	1	(H,Na)ClO ₄	lgK:1.35 (0.07MD)	3	EOD	20829
4	25	3	(H,Na)ClO ₄	lgK:2.54 (3SF)	0	MDS	140
5	25	3	NaClO ₄	lgK:2.54 (3SF)	0	MDS	437
6	20	4	HClO ₄	lgR:0.3 (0.07SD)	0	MDS	437
7	20	4	HClO ₄	lgR:0.5 (1SF)	0	MRX	437
8	20	4	HClO ₄	lgR:0.30 (2SF)	0	MCE	7033
9	20	4	HClO ₄	lgR:0.30 (2SF)	0	MDS	7033
10	20	4	HClO ₄	lgR:0.3 (0.4MD)	3	EOD	20829
11	20	4	HClO ₄	lgR:0.67 (0.01MD)	3	EOD	20829
12	25	0.5	HClO ₄	lgR:0.86 (2SF)	0	MCE	7033
13	25	0.96	HClO ₄	lgR:1.23 (3SF)	0	MCE	7033
14	25	2.88	HClO ₄	lgR:1.01 (3SF)	0	MCE	7033

Reaction No. 3084



No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	18:20	1	HCl	lgK:18.9 (0.07SD)	3	MLC	4433
2	19:21	2	HClO ₄	lgK:18.6 (3SF)	4	MIX	1118
3	20	0.23	HClO ₄	lgK:20.806 (5SF)	6	MIX	4405
4	20	1	HClO ₄	lgK:19.505 (5SF)	3	MIX	4405
5	20	1	NaClO ₄	lgK:20.2 (3SF)	5	EOD	1562
6	20	2	HClO ₄	lgK:18.59 (4SF)	5	MIX	13428
7	25	0	Inf. Dilution	lgK:24.1 (3SF)	3	CRV	552
8	25	0	Inf. Dilution	lgK:20.8 (3SF)	0	MGL	13427
9	25	0	Unknown	lgK:20.8 (0.1MD)	0	MIX	5106
10	25	0.1	KCl	lgK:20.8 (3SF)	0	CMN	1118
11	25	0	Inf. Dilution	dH:6.0 (2SF)	4	CRV	552

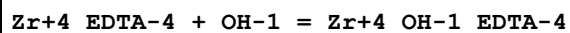
Reaction No. 4193



No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	19:21	2	HClO ₄	lgK:27.9 (3SF)	0	MIX	906
2	20	0.1	Unknown	lgK:29.5 (3SF)	4	ENS	773

3	20	0.23	HClO4	lgK:29.041 (5SF)	3	MIX	4405
4	20	1	HClO4	lgK:28.1 (3SF)	3	MIX	4405
5	20	1	HNO3	lgK:29. (0.3SD)	3	MIX	931
6	20	1	HNO3	lgK:28.46 (4SF)	5	MIX	931
7	20	1	NaClO4	lgK:27.7 (3SF)	4	MRX	142
8	20	1	Unknown	lgK:28. (0.4SD)	4	EIU	903
9	20	1.2	HCl	lgK:29.0 (0.04SD)	3	MIX	138
10	20	2	HClO4	lgK:27.91 (4SF)	0	MIX	13428
11	20	2	HNO3	lgK:30.58 (4SF)	0	MIX	931
Note (11): Low wgt for internal consistency							
12	20	3	HNO3	lgK:31.11 (4SF)	0	MIX	931
Note (12): Low wgt for internal consistency							
13	20	4	HNO3	lgK:30.92 (4SF)	0	MIX	931
14	20	5	HNO3	lgK:30.63 (4SF)	3	MIX	931
15	20	5	HNO3	lgK:30.6 (0.2SD)	0	MIX	931
16	23	3.6	HNO3	lgK:28.4 (3SF)	0	MPL	906
Note (16): Low wgt for internal consistency							
17	23			lgK:28.4 (3SF)	0	MPL	906
18	25	0	Inf. Dilution	lgK:33.25 (4SF)	1	EOR	
19	25	0	Inf. Dilution	lgK:32.8 (3SF)	0	CRV	552
Note (19): Low wgt for internal consistency							
20	25	0.1	KCl	lgK:29.9 (3SF)	3	MSP	140
21	25	0.1	KCl	lgK:19.9 (3SF)	0	MSP	13427
22	25	0.1	KNO3	lgK:29. (0.9SD)	0	MSP	903
Note (22): Low wgt for internal consistency							
23	25	0.1	NaClO4	lgK:19.40 (4SF)	0	MGL	140
24	25	0.1	Unknown	lgK:29.4 (3SF)	4	ENS	773
25	25	0	Unknown	dG:-44.73 (0.07MD)	2	MCL	5106
26	25	0	Inf. Dilution	dH:-0.6 (1SF)	4	CRV	552
27	25	0	Unknown	dH:-0.62 (0.15MD)	2	MCL	5106

Reaction No. 4194



No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	20	1	NaClO4	lgK:7.9 (2SF)	4	MRX	999
2	25	0.1	Unknown	lgK:7.6 (2SF)	4	ENS	773

Reaction No. 4255							
$2\text{Zr}+4_ \text{EDTA}-4 = \text{Zr}+4(2)_ \text{EDTA}-4(2)$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0.1	Unknown	lgK:3.5 (2SF)	4	ENS	773

Reaction No. 4269							
$\text{Zr}+4 + \text{DTPA}-5 = \text{Zr}+4_ \text{DTPA}-5$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	20	0.1	Unknown	lgK:36.00 (4SF)	3	ENS	1539
2	20	0.23	HClO ₄	lgK:35.806 (5SF)	3	MIX	4405
3	20	0.39	HCl	lgK:34.0 (3SF)	3	MIX	138
4	20	0.39	Unknown	lgK:33.96 (0.02SD)	3	MSC	999
5	20	1	HClO ₄	lgK:33.699 (5SF)	3	MIX	4405
6	20	1	NaClO ₄	lgK:36.9 (3SF)	0	MRX	999
Note (6): Expediency (hard to tell if H ⁺ affecting the DTPA or OH ⁻ the Zr+4)							
7	23	1	HClO ₄	lgK:33.69 (4SF)	5	MIX	4405
8	25	0	Inf. Dilution	lgK:40.6 (3SF)	2	EAE	

Reaction No. 4270							
$\text{Zr}+4_ \text{DTPA}-5 + \text{OH}-1 = \text{Zr}+4_ \text{OH}-1_ \text{DTPA}-5$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	20	1	NaClO ₄	lgK:8.1 (2SF)	4	MRX	999
2	20	1	NaClO ₄	lgK:8.1 (2SF)	5	CRV	13242
3	25	0	Inf. Dilution	lgK:7.8 (3SF)	2	EAE	

Reaction No. 4369							
$\text{Zr}+4 + \text{H}+1(2)_ \text{TTHA}-6 = \text{Zr}+4_ \text{H}+1(2)_ \text{TTHA}-6$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	20	0.5	Unknown	lgK:19.7 (3SF)	4	ENS	773

Reaction No. 4539							
$\text{Acac}-1 + \text{Zr}+4 = \text{Zr}+4_ \text{Acac}-1$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	18:20	1	HCl	lgK:11.255 (5SF)	6	MLC	4433
2	25	0	Inf. Dilution	lgK:11.3 (3SF)	1	ESC	

Reaction No. 4907							
$\text{Zr}^{+4} + 4\text{e}^{-1} = \text{Zr(s)}$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	$\lg K: -92.0 (3\text{SF})$	1	ELC	
2	25	0	Inf. Dilution	$\lg K: -101.5 (0.1\text{SD})$	0	ENS	5458
3	25	0	Inf. Dilution	$\lg K: -101.46 (5\text{SF})$	0	CRV	32224
4	25	0	Inf. Dilution	$\text{dG}: 522. (3\text{SF}) \text{ kJ}$	2	CRV	7421
5	25	0	Inf. Dilution	$\text{dG}: 133.270 (6\text{SF})$	0	CRV	9029
6	25	0	Inf. Dilution	$\text{dG}: 528.510 (9.227\text{MD}) \text{ kJ}$	2	CRV	20829
7	25	0	Inf. Dilution	$E: -1.45 (3\text{SF})$	0	CRV	6330
Note (7): Low wgt to achieve balance							
8	25	0	Inf. Dilution	$E: -1.45 (3\text{SF})$	0	CRV	7761
9	25	0	Inf. Dilution	$E: -1.45 (3\text{SF})$	0	CRV	22519
10	25	0	Inf. Dilution	$E: -1.55 (3\text{SF})$	0	CRV	22551
11	25	0	Inf. Dilution	$\text{dH}: 636.7 (4\text{SF}) \text{ kJ}$	0	CRV	7761
12	25	0	Inf. Dilution	$\text{dH}: 150.330 (6\text{SF})$	1	CRV	9029
13	25	0	Inf. Dilution	$\text{dH}: 608.500 (5.0\text{MD}) \text{ kJ}$	2	CRV	20829

Reaction No. 5015							
$\text{Zr}^{+4} + 2\text{CO}_3^{-2} + 2\text{OH}^{-1} = \text{Zr}^{+4}\text{OH-1(2)}\text{CO}_3^{-2(2)}$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	$\lg K: 46.9 (0.1\text{SD})$	4	MSL	29848

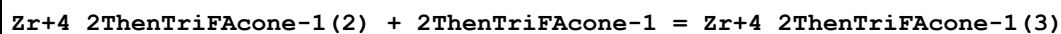
Reaction No. 9197							
$\text{Zr}^{+4} + \text{H+1(3)}_{\text{NTA-3}} = \text{Zr}^{+4}_{\text{NTA-3}} + 3\text{H+1}$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	23	1	HClO4	$\lg R: 5.4 (2\text{SF})$	3	MCE	22620
2	25	1	HClO4	$\lg R: 5.4 (0.05\text{SD})$	4	MIX	1118
3	25	2	HClO4	$\lg R: 4.1 (0.04\text{SD})$	4	MIX	1118

Reaction No. 13683							
$2\text{ThenTriFAcone-1} + \text{Zr}^{+4} = \text{Zr}^{+4}_{2\text{ThenTriFAcone-1}}$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	1	Unknown	$\lg K: 10.98 (4\text{SF})$	6	MDS	999

Reaction No. 13684							
$\text{Zr}^{+4}_{2\text{ThenTriFAcone-1}} + 2\text{ThenTriFAcone-1} = \text{Zr}^{+4}_{2\text{ThenTriFAcone-1(2)}}$							

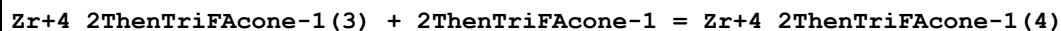
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	1	Unknown	lgK:10.90 (4SF)	6	MDS	999

Reaction No. 13685



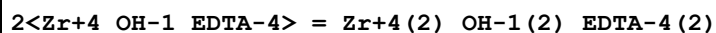
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	1	Unknown	lgK:10.36 (4SF)	6	MDS	999

Reaction No. 13686



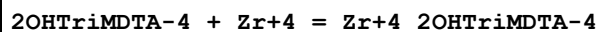
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	1	Unknown	lgK:9.93 (3SF)	6	MDS	999

Reaction No. 14070



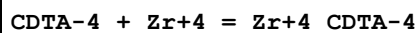
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0.1	KCl	lgK:3.5 (2SF)	4	MSP	140

Reaction No. 14445



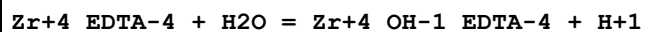
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	20	0.1	HCl	lgK:23.6 (0.04SD)	4	MSC	999

Reaction No. 14638



No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	10	0.1	KNO3	lgK:20.85 (4SF)	5	MEF	999
2	19:21	2	HClO4	lgK:29.9 (3SF)	0	MIX	999
3	20	0.1	KNO3	lgK:20.74 (4SF)	5	MEF	999
4	25	0	Inf. Dilution	lgK:30.0 (3SF)	1	ESC	
5	30	0.1	KNO3	lgK:20.64 (4SF)	5	MEF	999

Reaction No. 18084



No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	20	1	Unknown	lgK:-6.1 (2SF)	6	CRV	552
2	25	0.1	KCl	lgK:-6.2 (2SF)	6	MSP	999

Reaction No. 18117							
$\text{Zr+4_DTPA-5} + \text{H}_2\text{O} = \text{Zr+4_OH-1_DTPA-5} + \text{H+1}$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	20	0.1	Unknown	lgK:-5.90 (3SF)	6	ENS	1539
2	25	0	Inf. Dilution	lgK:-6.31 (3SF)	2	EAE	

Reaction No. 18183							
$\text{Zr+4} + \text{H+1_Citric-3} = \text{Zr+4_H+1_Citric-3}$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	18:20	1	HCl	lgK:10.778 (5SF)	2	MLC	4433
2	20	0.1	Unknown	lgK:10.80 (4SF)	0	ENS	1539
3	25	0	Inf. Dilution	lgK:8.52 (3SF)	2	CRV	31301

Reaction No. 18184							
$\text{Zr+4} + \text{H+1(2)_Citric-3} = \text{Zr+4_H+1(2)_Citric-3}$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	20	0.1	Unknown	lgK:6.20 (3SF)	6	ENS	1539
2	23	2	HClO4	lgK:3.41 (3SF)	5	MCE	3262
3	25	0	Inf. Dilution	lgK:7.16 (3SF)	2	EAE	

Reaction No. 18266							
$\text{Oxalic-2} + \text{Zr+4} = \text{Zr+4_Oxalic-2}$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	18:20	1	HCl	lgK:10.26 (0.05SD)	2	MLC	4433
2	20	0.1	Unknown	lgK:10.40 (4SF)	6	ENS	1539
3	20	1	HNO3	lgK:11.1 (0.2SD)	2	MIX	931
4	20	5	HNO3	lgK:11.3 (0.2SD)	3	MIX	931
5	23	2	HClO4	lgK:5.50 (3SF)	0	MCE	3262
6	23	2	HClO4	lgK:5.0 (2SF)	0	MCE	7744
7	25	0	Inf. Dilution	lgK:12.11 (4SF)	1	EOR	
8	25	0	Inf. Dilution	lgK:12.1 (3SF)	2	EAE	
9	25	0	Inf. Dilution	lgK:10.52 (4SF)	0	CRV	31301

Reaction No. 18267							
$2\text{<Oxalic-2>} + \text{Zr+4} = \text{Zr+4_Oxalic-2(2)}$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	20	0.1	Unknown	lgK:19.40 (4SF)	6	ENS	1539

2	20	1	KNO3	lgK:31.40 (4SF)	5	MIX	11516
Note (2): Caletka R et al., J. Inorg. Nucl. Chem., 1964, 26, 1443							
3	25	0	Inf. Dilution	lgK:20.42 (4SF)	1	EOR	
4	25	0	Inf. Dilution	lgK:20.4 (3SF)	2	EAE	
5	25	0	Inf. Dilution	lgK:18.15 (4SF)	0	CRV	31301

Reaction No. 18327							
Zr+4 + H+1_Tartaric-2 = Zr+4_H+1_Tartaric-2							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	18:20	1	HCl	lgK:6.9 (0.07SD)	4	MLC	4433
2	20	0.1	Unknown	lgK:6.00 (3SF)	6	ENS	1539
3	23	2	HClO4	lgK:2.49 (3SF)	0	MCE	3262

Reaction No. 18328							
Tiron-4 + Zr+4 = Zr+4_Tiron-4							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	20	0.1	Unknown	lgK:24.16 (4SF)	6	ENS	1539

Reaction No. 18523							
Zr+4 + H+1_Gly-1 = Zr+4_H+1_Gly-1							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0.1	Unknown	lgK:4.0 (0.09SD)	4	MGL	1555

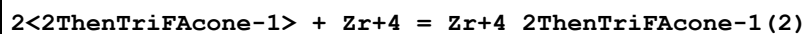
Reaction No. 19828							
Zr+4_Oxalic-2 + Oxalic-2 = Zr+4_Oxalic-2(2)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	20	1	HNO3	lgK:20.3 (0.3SD)	0	MIX	931
2	23	2	HClO4	lgK:9.68 (3SF)	0	MCE	3262
3	25	0	Inf. Dilution	lgK:8.3 (2SF)	1	ELC	

Reaction No. 20128							
Zr+4_DHEGly-1_OH-1 + H+1 = Zr+4_DHEGly-1 + H2O							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0.1	KNO3	lgK:3.9 (2SF)	4	MGL	999

Reaction No. 20129							
Zr+4_DHEGly-1_OH-1(2) + H+1 = Zr+4_DHEGly-1_OH-1 + H2O							

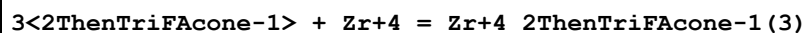
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0.1	KNO3	lgK:8.5 (2SF)	4	MGL	999

Reaction No. 20759



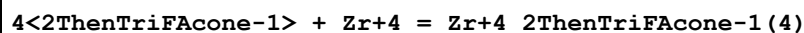
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	1	Unknown	lgK:21.88 (4SF)	6	ENS	773

Reaction No. 20760



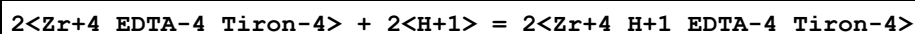
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	1	Unknown	lgK:32.24 (4SF)	6	ENS	773

Reaction No. 20761



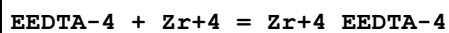
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	1	Unknown	lgK:42.17 (4SF)	6	ENS	773

Reaction No. 23024



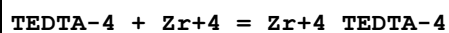
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0.1	Unknown	lgK:3.70 (3SF)	4	MGL	999

Reaction No. 24037



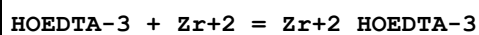
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	20	0.1	Unknown	lgK:24.72 (4SF)	6	MCG	999

Reaction No. 24057



No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	20	0.1	Unknown	lgK:23.17 (4SF)	6	MCG	999

Reaction No. 24364



No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0.1	KNO3	lgK:14.42 (4SF)	6	MGL	145

Reaction No. 29171							
Zr+4_DTPA-5 + H+1 = Zr+4_H+1_DTPA-5							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	20	0.1	Unknown	lgK:1.5 (2SF)	3	ENS	2373
2	25	0	Inf. Dilution	lgK:1.71 (3SF)	2	EAE	

Reaction No. 32312							
Zr+4_OH-1 + H+1_Mucic-2 = Zr+4_Mucic-2 + H2O							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	1	KCl	lgK:9.49 (3SF)	5	MSP	2261

Reaction No. 34302							
Zr+4_OH-1(2) + H+1_Chromotropic-4 = Zr+4_OH-1_Chromotropic-4 + H2O							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	16:23	0.1	KCl	lgK:18.68 (4SF)	5	MSP	931

Reaction No. 34591							
6<F-1> + Zr+4 = Zr+4_F-1(6)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	20	4	Unknown	lgK:38.4 (3SF)	0	CRV	2556
2	25	0	Inf. Dilution	lgK:38.110 (0.430MD)	3	CRV	20829
3	25	3	HClO4	lgK:12.62 (4SF)	0	ENS	437

Reaction No. 37309							
Salicylaldoxime-2 + Zr+4 = Zr+4_Salicylaldoxime-2							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:17.9 (3SF)	0	MGL	931
Note (1): Low wgt for internal consistency							
2	25	0.01	KCl	lgK:17.35 (4SF)	0	MGL	931
3	25	0.025	KCl	lgK:16.45 (4SF)	0	MGL	931
4	25	0.05	KCl	lgK:15.13 (4SF)	0	MGL	931
5	25	0.075	KCl	lgK:13.90 (4SF)	0	MGL	931
6	25	0.1	KCl	lgK:12.43 (4SF)	3	MGL	931

Reaction No. 37463							
Benzoylacetone-1 + Zr+4 = Zr+4_Benzoylacetone-1							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #

1	25	1	Benzene Unknown	lgK:12.71(0.04SD)	5	MDS	5076
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Reaction No. 37464							
Zr+4_Benzoylacetone-1 + Benzoylacetone-1 = Zr+4_Benzoylacetone-1(2)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	1	Benzene Unknown	lgK:11.86(0.02SD)	5	MDS	5076

Reaction No. 37465							
Zr+4_Benzoylacetone-1(2) + Benzoylacetone-1 = Zr+4_Benzoylacetone-1(3)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	1	Benzene Unknown	lgK:11.34(0.04SD)	5	MDS	5076

Reaction No. 37466							
Zr+4_Benzoylacetone-1(3) + Benzoylacetone-1 = Zr+4_Benzoylacetone-1(4)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	1	Benzene Unknown	lgK:11.08(0.04SD)	5	MDS	5076

Reaction No. 38777							
Cys-2 + Zr+4 = Zr+4_Cys-2							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	15	0.1	KNO3	lgK:14.40(4SF)	0	MGL	905
2	30	0.1	KNO3	lgK:14.15(4SF)	0	MGL	905

Reaction No. 39154							
BenzHydroxam-1 + Zr+4 = Zr+4_BenzHydroxam-1							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	1	HClO4	lgK:12.43(4SF)	5	MDS	931
2	25	1	HClO4	lgK:12.431(5SF)	5	MPT	4210

Reaction No. 39155							
2<BenzHydroxam-1> + Zr+4 = Zr+4_BenzHydroxam-1(2)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	1	HClO4	lgK:24.08(4SF)	5	MDS	931
2	25	1	HClO4	lgK:24.079(5SF)	5	MPT	4210

Reaction No. 40327							
MeDiPhos-4 + ZrO+2 = ZrO+2_MeDiPhos-4							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0.1	KCl	lgK:13.13 (4SF)	5	MGL	931

Reaction No. 40328							
2<MeDiPhos-4> + ZrO+2 = ZrO+2_MeDiPhos-4 (2)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0.1	KCl	lgK:19.45 (4SF)	5	MGL	931

Reaction No. 40329							
ZrO+2 + H+1_MeDiPhos-4 = ZrO+2_H+1_MeDiPhos-4							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0.1	KCl	lgK:9.01 (3SF)	5	MGL	931

Reaction No. 40330							
ZrO+2 + 2<H+1_MeDiPhos-4> = ZrO+2_H+1 (2)_MeDiPhos-4 (2)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0.1	KCl	lgK:12.18 (4SF)	5	MGL	931

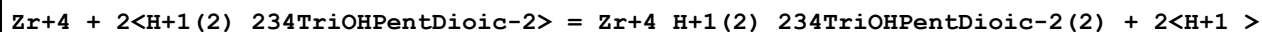
Reaction No. 40409							
Zr+4_Oxalic-2 (2) + Oxalic-2 = Zr+4_Oxalic-2 (3)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:4.0 (2SF)	1	ESC	
2	25			lgK:4.0 (0.1SD)	0	MGL	931

Reaction No. 40410							
Zr+4 + H+1 (2)_Oxalic-2 = Zr+4_Oxalic-2 + 2<H+1>							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:6.581 (4SF)	1	ELC	
2	23	2	HClO4	lgR:5.5 (0.04SD)	3	MIX	931
3	23	4	HClO4	lgR:5.6 (0.07SD)	3	MIX	931

Reaction No. 40917							
Zr+4 + H+1 (2)_234TriOHPentDioic-2 = Zr+4_H+1_234TriOHPentDioic-2 + H+1							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	23	2	HClO4	lgR:3.09 (3SF)	5	MIX	931

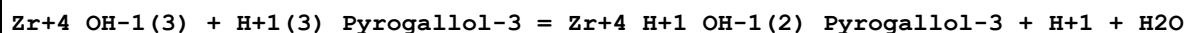
2	23	2	HClO4	lgR:3.41(0.01SD)	5	MIX	3262
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Reaction No. 40918



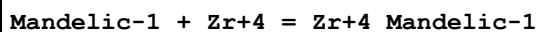
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	23	2	HClO4	lgR:4.14(3SF)	3	MIX	931
2	23	2	HClO4	lgR:5.4(0.07SD)	3	MIX	3262

Reaction No. 41690



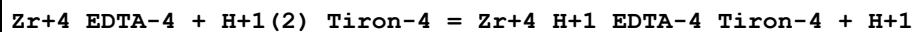
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	18:20	0	Inf. Dilution	lgK:4.17(3SF)	4	MSP	4406
2	18:20	0.1	KCl	lgK:4.06(3SF)	4	MSP	4406

Reaction No. 41727



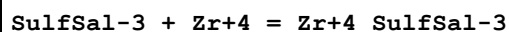
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	18:20	1	HCl	lgK:5.6(0.07SD)	5	MLC	4433

Reaction No. 41936



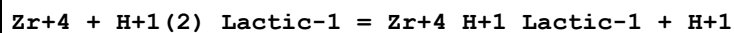
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0.1	KCl	lgK:-0.92(2SF)	3	MGL	13427
2	25	0.1	Unknown	lgK:-12.(2SF)	0	MGL	140

Reaction No. 42068



No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	18:20	1	HCl	lgK:4.792(4SF)	0	MCL	4433
2	25	0	Inf. Dilution	lgK:4.8(2SF)	1	ESC	

Reaction No. 42252



No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:2.28(3SF)	2	EAE	
2	23	2	HClO4	lgR:2.28(3SF)	5	MIX	3262

Reaction No. 42253

Zr+4 + 2<H+1(2)_Lactic-1> = Zr+4_H+1(2)_Lactic-1(2) + 2<H+1>							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:2.54(3SF)	2	EAE	
2	23	2	HClO4	lgR:2.54(3SF)	5	MIX	3262

Reaction No. 42255							
Zr+4 + H+1(2)_Malic-2 = Zr+4_H+1_Malic-2 + H+1							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:-0.2399(4SF)	1	ELC	
2	23	2	HClO4	lgR:2.24(3SF)	3	MIX	3262

Reaction No. 42257							
Zr+4 + H+1(2)_Tartaric-2 = Zr+4_H+1_Tartaric-2 + H+1							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	23	2	HClO4	lgR:2.49(3SF)	5	MIX	3262

Reaction No. 42259							
Zr+4 + H+1(2)_Citric-3 = Zr+4_H+1_Citric-3 + H+1							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:3.75(3SF)	1	ELC	
2	23	2	HClO4	lgR:3.42(3SF)	3	MIX	3262

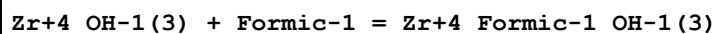
Reaction No. 42375							
Zr+4 + H+1(2)_Oxalic-2 = Zr+4_H+1_Oxalic-2 + H+1							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:5.23(3SF)	2	EAE	
2	23	2	HClO4	lgR:4.50(3SF)	4	MIX	3262

Reaction No. 42376							
Zr+4 + 2<H+1(2)_Oxalic-2> = Zr+4_H+1(2)_Oxalic-2(2) + 2<H+1>							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:9.53(3SF)	2	EAE	
2	23	2	HClO4	lgR:8.32(3SF)	4	MIX	3262

Reaction No. 42378							
Zr+4 + 4<H+1(2)_Oxalic-2> = Zr+4_H+1(4)_Oxalic-2(4) + 4<H+1>							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #

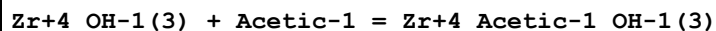
1	25	0	Inf. Dilution	lgK:5.85 (3SF)	2	EAE	
2	23	2	HClO4	lgR:4.397 (4SF)	4	MIX	3262

Reaction No. 42783



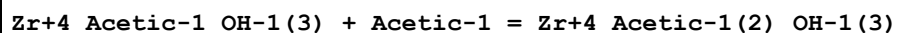
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0.01	Unknown	lgK:3.4 (0.04SD)	4	MHY	906

Reaction No. 42941



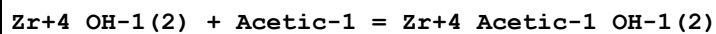
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0.01	Unknown	lgK:3.35 (3SF)	4	MHY	906

Reaction No. 42942



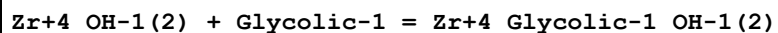
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0.01	Unknown	lgK:1.83 (3SF)	4	MHY	906

Reaction No. 42943



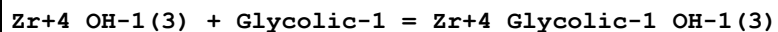
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:6.2 (0.05SD)	5	MKN	906

Reaction No. 42952



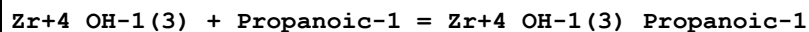
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	23	1	KCl	lgK:6.5 (0.03SD)	5	MSP	906
2	25	0	Inf. Dilution	lgK:7.7 (2SF)	5	MKN	906

Reaction No. 42953



No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	23	1	KCl	lgK:6.6 (0.03SD)	5	MSP	906

Reaction No. 43055



No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0.01	Unknown	lgK:3.8 (0.07SD)	5	MHY	906

Reaction No. 43270							
Zr+4_OH-1(3) + Tartaric-2 = Zr+4_OH-1(3)_Tartaric-2							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	15			lgK:6.24(3SF)	4	MHY	906
2	20			lgK:6.15(3SF)	4	MHY	906
3	25			lgK:6.07(3SF)	4	MHY	906
4	25			lgK:6.09(3SF)	4	MHY	906
5	30			lgK:5.92(3SF)	4	MHY	906
6	35			lgK:5.84(3SF)	4	MHY	906

Reaction No. 43271							
Zr+4_OH-1(3)_Tartaric-2 + Tartaric-2 = Zr+4_OH-1(3)_Tartaric-2(2)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25			lgK:3.06(3SF)	4	MHY	906

Reaction No. 43272							
Zr+4_OH-1(2) + H+1(2)_Tartaric-2 = Zr+4_H+1(2)_OH-1(2)_Tartaric-2							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	23			lgK:9.8(0.03SD)	4	MIX	906

Reaction No. 43276							
Zr+4_OH-1(3) + Butanoic-1 = Zr+4_Butanoic-1_OH-1(3)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0.01	Unknown	lgK:3.8(0.06SD)	4	MHY	906

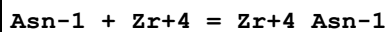
Reaction No. 43356							
Zr+4 + Asp-2 = Zr+4_Asp-2							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0.1	NaClO4	lgK:9.70(3SF)	5	MGL	22615

Reaction No. 43357							
Zr+4_Asp-2 + Asp-2 = Zr+4_Asp-2(2)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0.1	NaClO4	lgK:6.85(3SF)	4	MGL	22615

Reaction No. 43358							
Zr+4_Asp-2(2) + Asp-2 = Zr+4_Asp-2(3)							

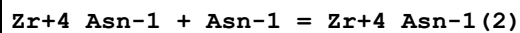
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0.1	NaClO4	lgK:3.50 (3SF)	4	MGL	22615

Reaction No. 43425



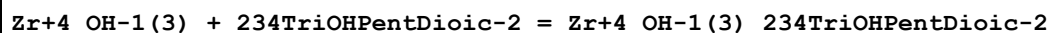
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0.1	NaClO4	lgK:8.80 (3SF)	4	MGL	906

Reaction No. 43426



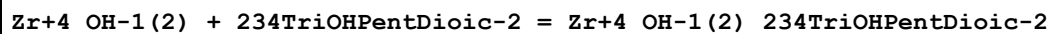
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0.1	NaClO4	lgK:6.25 (3SF)	4	MGL	906

Reaction No. 43584



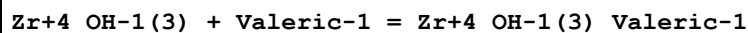
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25			lgK:6.86 (3SF)	4	MHY	906

Reaction No. 43585



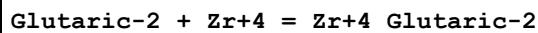
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	23	2.25	NaCl	lgK:11.4 (0.1SD)	4	MIX	906

Reaction No. 43589



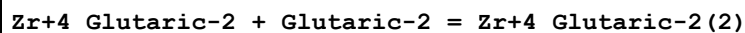
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0.01	Unknown	lgK:3.9 (0.09SD)	4	MHY	906

Reaction No. 43628



No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0.1	NaClO4	lgK:9.60 (3SF)	4	MGL	906

Reaction No. 43629



No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0.1	NaClO4	lgK:6.40 (3SF)	4	MGL	906

Reaction No. 43630							
Zr+4_Glutaric-2(2) + Glutaric-2 = Zr+4_Glutaric-2(3)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0.1	NaClO4	lgK:3.32 (3SF)	4	MGL	906

Reaction No. 43639							
Gln-1 + Zr+4 = Zr+4_Gln-1							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0.1	NaClO4	lgK:8.75 (3SF)	4	MGL	906

Reaction No. 43640							
Zr+4_Gln-1 + Gln-1 = Zr+4_Gln-1(2)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0.1	NaClO4	lgK:6.10 (3SF)	4	MGL	906

Reaction No. 43705							
Zr+4 + H+1_Ascorbic-2 = Zr+4_H+1_Ascorbic-2							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	23			lgK:9.5 (2SF)	4	MSP	906

Reaction No. 43768							
Zr+4_OH-1(3) + Mucic-2 = Zr+4_OH-1(3)_Mucic-2							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25			lgK:6.55 (3SF)	4	MHY	906

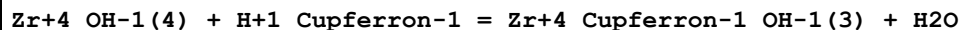
Reaction No. 43776							
2<Gluconic-1> + ZrO+2 = ZrO+2_Gluconic-1(2)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	22	0.2	NaClO4	lgK:2.25 (3SF)	4	MSP	906

Reaction No. 43777							
Zr+2_OH-1(2) + Gluconic-1 = Zr+2_Gluconic-1_OH-1(2)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	23	0.1	HCl	lgK:6.60 (3SF)	4	MIX	906
2	23	2	NaCl	lgK:6.60 (3SF)	4	MIX	906

Reaction No. 43778							
Zr+2_Gluconic-1_OH-1(2) + Gluconic-1 = Zr+2_Gluconic-1(2)_OH-1(2)							

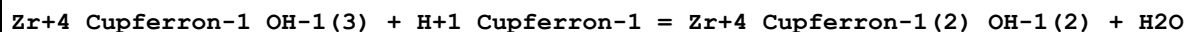
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	23	0.1	HCl	lgK:5.95(3SF)	4	MIX	906
2	23	2	NaCl	lgK:5.95(3SF)	4	MIX	906

Reaction No. 43986



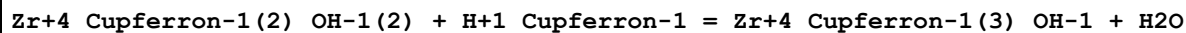
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	23		Ethanol:Water 40:60Vol	lgK:7.36(3SF)	4	MSP	906

Reaction No. 43987



No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	23		Ethanol:Water 40:60Vol	lgK:6.74(3SF)	4	MSP	906

Reaction No. 43988



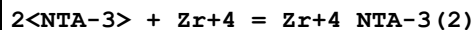
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	23		Ethanol:Water 40:60Vol	lgK:6.21(3SF)	4	MSP	906

Reaction No. 43989



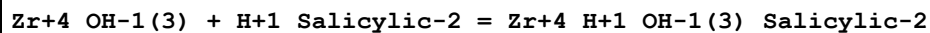
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	23		Ethanol:Water 40:60Vol	lgK:5.84(3SF)	4	MSP	906

Reaction No. 44022



No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	23	2	HClO4	lgK:7.8(0.2SD)	0	MSP	906
2	25	0	Inf. Dilution	lgK:32.5(3SF)	1	ELC	

Reaction No. 44202



No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	23			lgK:6.38(0.01SD)	4	MSP	906
2	25			lgK:6.1(0.2SD)	4	MKN	906

Reaction No. 44203							
$\text{Zr+4_OH-1(2)} + \text{H+1_Salicylic-2} = \text{Zr+4_H+1_OH-1(2)_Salicylic-2}$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	23			lgK:6.0(0.07SD)	4	MSP	906

Reaction No. 44344							
$\text{Zr+4_OH-1(3)} + \text{H+1_SulfSal-3} = \text{Zr+4_H+1_OH-1(3)_SulfSal-3}$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	23	1	HCl	lgK:6.2(0.2SD)	4	MSP	906
2	23	1	KCl	lgK:6.2(0.2SD)	4	MSP	906
3	25	0	Inf. Dilution	lgK:6.2(2SF)	1	ESC	
4	25			lgK:6.0(0.2SD)	0	MKN	906

Reaction No. 44345							
$\text{Zr+4_OH-1(2)} + \text{H+1_SulfSal-3} = \text{Zr+4_H+1_OH-1(2)_SulfSal-3}$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	23	1	HCl	lgK:6.2(0.06SD)	4	MSP	906
2	23	1	KCl	lgK:6.2(0.06SD)	4	MSP	906
3	25	0	Inf. Dilution	lgK:6.2(2SF)	1	ESC	

Reaction No. 44455							
$\text{AmPhenMeDiPhos-4} + \text{ZrO+2} = \text{ZrO+2_AmPhenMeDiPhos-4}$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0.1	KCl	lgK:17.08(4SF)	5	MGL	906

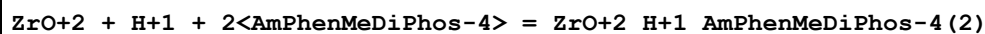
Reaction No. 44456							
$2<\text{AmPhenMeDiPhos-4}> + \text{ZrO+2} = \text{ZrO+2_AmPhenMeDiPhos-4(2)}$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0.1	KCl	lgK:21.66(4SF)	5	MGL	906

Reaction No. 44457							
$\text{ZrO+2} + \text{H+1_AmPhenMeDiPhos-4} = \text{ZrO+2_H+1_AmPhenMeDiPhos-4}$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0.1	KCl	lgK:11.61(4SF)	5	MGL	906

Reaction No. 44458							
$2<\text{ZrO+2}> + \text{AmPhenMeDiPhos-4} = \text{ZrO+2(2)_AmPhenMeDiPhos-4}$							

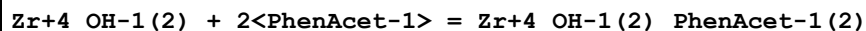
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0.1	KCl	lgK:23.18 (4SF)	5	MGL	906

Reaction No. 44459



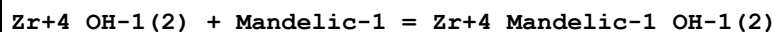
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0.1	KCl	lgK:16.69 (4SF)	5	MGL	906

Reaction No. 44469



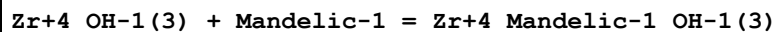
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:6.6 (2SF)	4	MKN	906
2	25	0.01	Unknown	lgK:6.8 (2SF)	4	MKN	906
3	25	0.1	Unknown	lgK:6.2 (2SF)	4	MKN	906

Reaction No. 44486



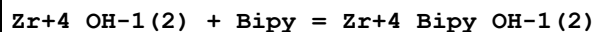
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	23	1	KCl	lgK:6.7 (0.04SD)	4	MSP	906

Reaction No. 44487



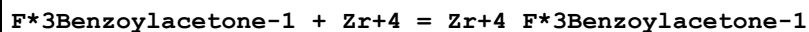
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	23	1	KCl	lgK:6.71 (0.02SD)	4	MSP	906

Reaction No. 44739



No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	20	0.1	Unknown	lgK:9.7 (0.03SD)	4	MSP	906

Reaction No. 44782



No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	1	Unknown	lgK:10.85 (4SF)	4	MDS	906

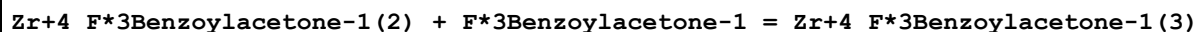
Reaction No. 44783



No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
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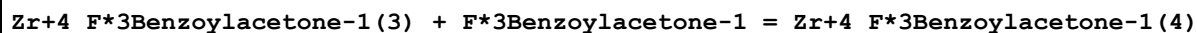
1	25	1	Unknown	lgK:10.35 (4SF)	4	MDS	906
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Reaction No. 44784



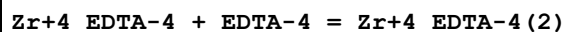
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	1	Unknown	lgK:9.85 (3SF)	4	MDS	906

Reaction No. 44785



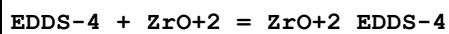
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	1	Unknown	lgK:9.35 (3SF)	4	MDS	906

Reaction No. 44903



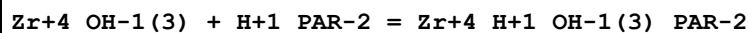
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	23			lgK:8. (0.3SD)	4	MSP	906
2	25	0	Inf. Dilution	lgK:8.0 (2SF)	1	ESC	

Reaction No. 44913



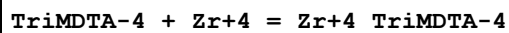
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	30	0.1	KNO3	lgK:11.20 (4SF)	4	MGL	906

Reaction No. 45069



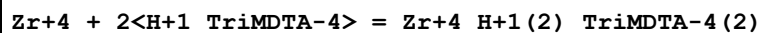
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	20	0.1	Unknown	lgK:16.44 (4SF)	4	MSP	906

Reaction No. 45087



No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	23	1	HClO4	lgK:28.33 (4SF)	4	MDS	906

Reaction No. 45088



No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	23	1	HClO4	lgK:33.02 (4SF)	4	MDS	906

Reaction No. 45111							
$\text{Zr+4_OH-1(2)} + 2\langle\text{Phenanth}\rangle = \text{Zr+4_Phenanth(2)_OH-1(2)}$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	20	0.1:1	Unknown	lgK:10.07 (0.01SD)	4	MSP	906

Reaction No. 45131							
$\text{EDDG-4} + \text{ZrO+2} = \text{ZrO+2_EDDG-4}$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	30	0.1	KNO3	lgK:11.80 (4SF)	4	MGL	906

Reaction No. 45283							
$\text{Zr+4} + \text{H+1(5)_DTPA-5} = \text{Zr+4_DTPA-5} + 5\langle\text{H+1}\rangle$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	23			lgK:4.43 (3SF)	0	MSP	906
2	25	0	Inf. Dilution	lgK:9.9 (2SF)	1	ELC	

Reaction No. 46382							
$\text{Zr+4_Oxalic-2} + \text{H+1} + \text{H2O} = \text{Zr+4_OH-1} + \text{H+1(2)_Oxalic-2}$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:-6.265 (4SF)	1	ELC	
2	25	4	NaClO4	lgK:4.30 (3SF)	0	MDS	4708

Reaction No. 52453							
$\text{Zr+4_Cl-1(4)} + 2\langle\text{2NitrAniline}\rangle = \text{Zr+4_2NitrAniline(2)_Cl-1(4)}$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25		Dioxan	lgK:4.78 (3SF)	4	MSP	4695

Reaction No. 52454							
$\text{Zr+4_Cl-1(4)} + 2\langle\text{35DiNitrAniline}\rangle = \text{Zr+4_35DiNitrAniline(2)_Cl-1(4)}$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25		Dioxan	lgK:4.15 (3SF)	4	MSP	4695

Reaction No. 52455							
$\text{Zr+4_Cl-1(4)} + 2\langle\text{2Me4NitrAniline}\rangle = \text{Zr+4_2Me4NitrAniline(2)_Cl-1(4)}$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25		Dioxan	lgK:5.58 (3SF)	4	MSP	4695

Reaction No. 52994							
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Cl-1 + Zr+4 = Zr+4_Cl-1							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	18:30	2	HClO4	lgK:0.03 (1SF)	6	CRV	20829
2	18:30	2	HClO4	lgK:-0.02 (1SF)	4	MIX	22619
3	18:30	4	HClO4	lgK:0.05 (1SF)	6	CRV	20829
4	18:30	4	HClO4	lgK:0.04 (1SF)	4	MIX	22619
5	20	2	HClO4	lgK:0.08 (1SF)	5	MDS	5398
6	20	2	HClO4	lgK:0.079 (2SF)	4	MIX	13428
7	20	3.5	HClO4	lgK:-0.5 (1SF)	0	MSP	5398
8	20	6.54	H(ClO4)	lgK:0.92 (2SF)	3	MDS	140
9	23	2	HClO4	lgK:-0.02 (1SF)	3	MIX	3262
10	23	4	H(ClO4)	lgK:0.04 (1SF)	3	MCE	140
11	23	4	HClO4	lgK:0.04 (1SF)	3	MIX	3262
12	25	0	Inf. Dilution	lgK:0.3 (0.05SD)	0	ENS	5458
13	25	0	Inf. Dilution	lgK:0.19 (0.11SD)	2	EOD	7033
14	25	0	Inf. Dilution	lgK:1.590 (0.060MD)	6	CRV	20829
15	25	0	Inf. Dilution	lgK:0.50 (2SF)	4	CRV	32224
16	25	2	HClO4	lgK:0.30 (2SF)	3	MDS	140
17	25	2	HClO4	lgK:0.30 (2SF)	2	MSE	7998
18	25	2	HClO4	lgK:-0.02 (1SF)	5	MIX	22619
19	25	6.5	HClO4	lgK:0.92 (2SF)	4	MIX	5502
20	25	6.54	HCl	lgR:0.92 (0.05SD)	5	MDS	5502

Reaction No. 52995							
Zr+4 + 2<Cl-1> = Zr+4_Cl-1(2)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	20	2	HClO4	lgK:-0.54 (2SF)	0	MDS	5398
2	23	4	H(ClO4)	lgK:-0.68 (2SF)	1	MCE	140
3	25	0	Inf. Dilution	lgK:0.48 (2SF)	1	ELC	
4	25	0	Inf. Dilution	lgK:0.48 (2SF)	1	EOR	
5	25	0	Inf. Dilution	lgK:2.170 (0.240MD)	0	CRV	20829
6	25	6.54	HCl	lgR:1.32 (0.05SD)	2	MDS	5502

Reaction No. 52996							
Zr+4 + 3<Cl-1> = Zr+4_Cl-1(3)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #

1	20	2	HClO4	lgK:-1.0(2SF)	0	MDS	5398
2	23	4	H(ClO4)	lgK:-1.30(3SF)	0	MCE	140
3	25	0	Inf. Dilution	lgK:-0.6(1SF)	1	ELC	
4	25	6.54	HCl	lgR:1.51(0.05SD)	0	MDS	5502

Reaction No. 53080							
5<CO3-2> + Zr+4 = Zr+4_CO3-2(5)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:41.6(0.2SD)	4	MSL	29848
2	25	0	Inf. Dilution	lgK:43.(2SF)	2	EOD	29848

Reaction No. 53160							
Zr+4_F-1(5) + F-1 = Zr+4_F-1(6)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	23	0.5	HCl	lgK:3.6(2SF)	0	MEF	5398
2	23			lgK:3.84(3SF)	0	MIX	5398
3	25	0	Inf. Dilution	lgK:3.3(3SF)	0	EAE	
4	25	0	Inf. Dilution	lgK:0.75(0.38SD)	0	EOD	7033
5	25	1	NaClO4	lgK:1.4(0.05SD)	0	MQH	931
Note (5): Low wgt for internal consistency (not in accord with no-I data)							
6	25	0	Inf. Dilution	dH:4.0(5.SD)	0	MCL	437
7	25			dH:3.800(4SF)	0	MCL	437

Reaction No. 53161							
Zr+4_F-1(2) + 2<H+1> = Zr+4 + 2<H+1_F-1>							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:-5.7(2SF)	0	ELC	
2	25	1	HClO4	lgR:-9.11(3SF)	0	MCE	5398
3	25	1	HNO3	lgR:-9.65(3SF)	0	MCE	5398
4	25	2	HClO4	lgR:-9.38(3SF)	0	MCE	5398
5	25	2	HNO3	lgR:-9.53(3SF)	0	MCE	5398

Reaction No. 53302							
Zr+4 + H+1_SO4-2 = Zr+4_SO4-2 + H+1							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:4.6(2SF)	1	ELC	
2	25	0	Inf. Dilution	lgK:2.6(2SF)	0	EDH	29883

Note (2): K > 4.6E2							
3	20	2	HClO4	lgR:2.6 (2SF)	5	MDS	5398
4	20	2	HClO4	lgR:2.6 (2SF)	4	MIX	13428
5	20	4	HClO4	lgR:2.85 (3SF)	3	MCE	140
6	20	4	HClO4	lgR:2.7 (2SF)	3	MSE	13458
7	20	4	HClO4	lgR:2.8 (2SF)	3	MCE	22616
8	23	2.3	HClO4	lgR:2.56 (3SF)	4	MCE	140
9	23	2.3	HClO4	lgR:2.67 (3SF)	5	MCE	931
10	25	2	HClO4	lgR:2.66 (3SF)	5	MDS	140

Reaction No. 53303							
Zr+4_SO4-2 + H+1_SO4-2 = Zr+4_SO4-2(2) + H+1							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:3.2 (2SF)	1	ELC	
2	25	0	Inf. Dilution	lgK:1.72 (3SF)	0	EDH	29883
3	20	2	HClO4	lgR:1.9 (2SF)	4	MIX	13428
4	20	4	HClO4	lgR:1.85 (3SF)	5	MCE	140
5	20	4	HClO4	lgR:0.9 (1SF)	3	MDS	5398
6	20	4	HClO4	lgR:1.9 (2SF)	4	MSE	13458
7	20	4	HClO4	lgR:1.85 (3SF)	4	MCE	22616
8	25	2	HClO4	lgR:1.72 (3SF)	5	MDS	140

Reaction No. 53304							
Zr+4_SO4-2(2) + H+1_SO4-2 = Zr+4_SO4-2(3) + H+1							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:-0.24 (2SF)	1	ELC	
2	25	0	Inf. Dilution	lgK:-0.2 (1SF)	0	ELC	
3	25	0	Inf. Dilution	lgK:0.0 (1SF)	0	EDH	29883
4	20	2	HClO4	lgR:1.0 (2SF)	2	MIX	13428
5	20	4	HClO4	lgR:0.3 (1SF)	3	MDS	5398
6	25	2	HClO4	lgR:0.0 (1SF)	3	MDS	140

Reaction No. 53305							
Zr+4 + 2<H+1_SO4-2> = Zr+4_SO4-2(2) + 2<H+1>							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:8.7 (2SF)	1	ELC	
2	20	2	HClO4	lgR:4.5 (2SF)	3	MDS	5398

Note (2): Low wgt for internal consistency							
3	23	2.3	HClO4	lgR:3.34 (3SF)	3	MCE	140
4	23	2.3	HClO4	lgR:3.54 (3SF)	3	MCE	931

Reaction No. 53306							
Zr+4 + 3<H+1_SO4-2> = Zr+4_SO4-2(3) + 3<H+1>							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:8.4 (2SF)	1	ELC	
2	20	2	HClO4	lgR:5.5 (2SF)	3	MDS	5398
3	23	2.3	HClO4	lgR:5.61 (3SF)	3	MCE	140
4	23	2.3	HClO4	lgR:5.59 (3SF)	3	MCE	931

Reaction No. 53370							
Zr+4 + 3<H+1_PO4-3> = Zr+4_H+1(3)_PO4-3(3)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	20	1	NaClO4	lgK:1.59 (3SF)	5	MSP	5398
2	25	0	Inf. Dilution	lgK:5.14 (3SF)	2	EAE	

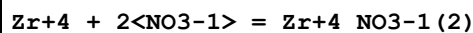
Reaction No. 53503							
Zr+4 + H2O2 = Zr+4_H2O2							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	20	1	HCl	lgK:3.10 (3SF)	3	MSP	5398
2	25	0	Inf. Dilution	lgK:3.1 (3SF)	2	EAE	

Reaction No. 53504							
ZrO+2 + H2O2 = ZrO+2_H2O2							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25			lgK:4.0 (2SF)	0	MDG	5398

Reaction No. 53656							
Zr+4 + NO3-1 = Zr+4_NO3-1							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	18:30	2	HClO4	lgK:0.05 (1SF)	6	CRV	20829
2	18:30	2	HClO4	lgK:-0.03 (1SF)	4	MIX	22619
3	18:30	4	HClO4	lgK:-0.09 (1SF)	6	CRV	20829
4	18:30	4	HClO4	lgK:-0.04 (1SF)	4	MIX	22619
5	20	2	HClO4	lgK:-0.05 (1SF)	4	MDS	5398

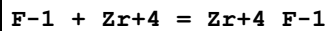
6	20	2	HClO4	lgK:-0.04 (1SF)	4	MIX	13428
7	20	4	HClO4	lgK:0.34 (2SF)	3	MDS	140
8	23	2	HClO4	lgK:-0.04 (1SF)	4	MCE	140
9	23	2	HClO4	lgK:-0.03 (1SF)	3	MIX	3262
10	23	4	HClO4	lgK:-0.06 (1SF)	4	MCE	140
11	23	4	HClO4	lgK:-0.06 (1SF)	3	MIX	3262
12	25	0	Inf. Dilution	lgK:0.3 (0.05SD)	0	ENS	5458
13	25	0	Inf. Dilution	lgK:1.590 (0.080MD)	6	CRV	20829
14	25	0	Inf. Dilution	lgK:0.3 (1SF)	0	CRV	32224
15	25	2	HClO4	lgK:0.3 (1SF)	2	MDS	140
16	25	2	HClO4	lgK:0.30 (2SF)	2	MSE	7998
17	25	2	HClO4	lgK:-0.05 (1SF)	5	MIX	13428
18	25	4	HClO4	lgK:0.34 (2SF)	2	MIX	5502
19	25	4	HNO3	lgR:0.34 (0.05SD)	5	MDS	5502

Reaction No. 53657



No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	20	2	HClO4	lgK:-0.46 (2SF)	2	MDS	5398
2	23	2	HClO4	lgK:-0.34 (2SF)	0	MCE	140
3	23	2	HClO4	lgK:0.22 (2SF)	0	MCE	140
4	23	4	HClO4	lgK:-0.85 (2SF)	0	MCE	140
5	25	0	Inf. Dilution	lgK:3.3 (2SF)	1	ELC	
6	25	0	Inf. Dilution	lgK:2.640 (0.170MD)	0	CRV	20829
7	25	4	HNO3	lgR:0.11 (0.05SD)	3	MDS	5502

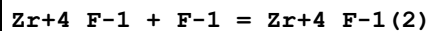
Reaction No. 54492



No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:6.4 (2SF)	0	ELC	
2	25	0	Inf. Dilution	lgK:9.8 (0.05SD)	0	ENS	5458
3	25	0	Inf. Dilution	lgK:5.84 (0.07SD)	0	EOD	7033
4	25	0	Inf. Dilution	lgK:10.120 (0.070MD)	3	CRV	20829
5	25	0	Inf. Dilution	lgK:9.8 (2SF)	0	CRV	32224
6	25	2	Unknown	lgK:8.94 (3SF)	0	CRV	2556
7	25	4	Unknown	lgK:9.44 (3SF)	0	CRV	2556

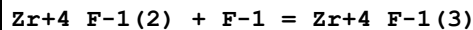
8	25	4	HClO ₄	dG:-54.1 (3SF) kJ	0	MMX	4923
9	25	0	Inf. Dilution	dH:-17.500 (0.700MD) kJ	3	CRV	20829
10	25	4	HClO ₄	dH:-5.6 (2SF) kJ	0	MMX	4923
11	25	4	Unknown	dH:-1.3 (2SF)	0	CRV	2556
12	25			dH:5.975 (4SF)	0	MCL	437

Reaction No. 54493



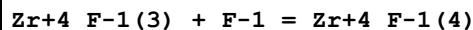
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:8.4 (2SF)	1	ELC	
2	25	0	Inf. Dilution	lgK:4.39 (0.15SD)	0	EOD	7033
3	25	4	HClO ₄	dG:-44.9 (3SF) kJ	2	MMX	4923
4	25	4	HClO ₄	dH:-4.9 (2SF) kJ	2	MMX	4923
5	25			dH:5.975 (4SF)	0	MCL	437

Reaction No. 54494



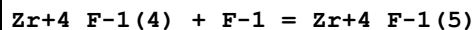
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:6.1 (2SF)	1	ELC	
2	25	0	Inf. Dilution	lgK:2.89 (0.16SD)	0	EOD	7033
3	25	4	HClO ₄	dG:-37.1 (3SF) kJ	0	MMX	4923
4	25	4	HClO ₄	dH:0.7 (1SF) kJ	0	MMX	4923
5	25			dH:5.975 (4SF)	0	MCL	437

Reaction No. 54495



No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:3.7 (2SF)	0	ELC	
2	25	0	Inf. Dilution	lgK:2.18 (0.26SD)	0	EOD	7033
3	25	1	NaClO ₄	lgK:2.8 (0.05SD)	0	MQH	931
4	25	4	HClO ₄	dG:-32.8 (3SF) kJ	0	MMX	4923
5	25	0	Inf. Dilution	dH:10. (5.SD)	0	MCL	437
6	25	4	HClO ₄	dH:-10. (2SF) kJ	0	MMX	4923
7	25			dH:12.069 (5SF)	0	MCL	437

Reaction No. 54496



No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:3.9(2SF)	1	ELC	
2	25	0	Inf. Dilution	lgK:1.51(0.06SD)	0	EOD	7033
3	25	1	NaClO ₄	lgK:1.9(0.05SD)	0	MQH	931
4	25	0	Inf. Dilution	dH:-4.0(5.SD)	0	MCL	437
5	25			dH:4.302(4SF)	0	MCL	437

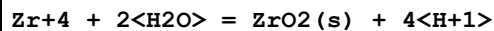
Reaction No. 55129							
SO ₄ -2 + Zr+4 = Zr+4_SO ₄ -2							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	20	2	HClO ₄	lgK:2.63(3SF)	5	MIX	13428
2	20	2	HClO ₄	lgK:2.56(3SF)	5	MSC	19258
3	20	4	HClO ₄	lgK:2.78(3SF)	3	MSE	13458
4	20	4	Unknown	lgK:3.68(3SF)	3	CRV	2556
5	23	2	HClO ₄	lgK:2.67(3SF)	3	MIX	3262
6	25	0	Inf. Dilution	lgK:3.8(0.05SD)	0	ENS	5458
7	25	0	Inf. Dilution	lgK:7.040(0.090MD)	3	CRV	20829
8	25	0	Inf. Dilution	lgK:3.61(3SF)	0	EOD	22958
9	25	0	Inf. Dilution	lgK:7.04(0.09SD)	3	EOD	31756
10	25	2	HClO ₄	lgK:3.74(3SF)	0	MDS	140
11	25	2	HClO ₄	lgK:2.66(3SF)	5	MSE	7998
12	25	2	HClO ₄	lgK:2.63(3SF)	5	MSE	13652
13	25	2	NaClO ₄	lgK:3.79(3SF)	0	MDS	8136
14	25	2	Unknown	lgK:3.67(3SF)	3	CRV	2556
15	25	4	HClO ₄	lgK:2.85(3SF)	5	MCE	22616
16	25	4	NaClO ₄	lgK:1.60(3SF)	0	MDS	4708
17	25			lgK:3.79(3SF)	4	CRV	32224

Reaction No. 55130							
Zr+4 + 4<F-1> = Zr+4_F-1(4)_(s)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:27.2(3SF)	0	ELC	
2	25	0	Inf. Dilution	lgK:18.7(0.05SD)	0	ENS	5458
3	25	0	Inf. Dilution	lgK:18.72(4SF)	0	CRV	32224

Reaction No. 55131							
Zr+4 + 4<Cl-1> = Zr+4_Cl-1(4)_(s)							

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:-28.8(3SF)	1	ELC	
2	25	0	Inf. Dilution	lgK:-37.5(0.05SD)	0	ENS	5458
3	25	0	Inf. Dilution	lgK:-37.53(4SF)	0	CRV	32224
4	25	0	Inf. Dilution	dH:-70.(2.SD)	3	MCL	5106

Reaction No. 55132

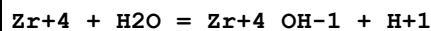


No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:7.4(2SF)	1	ELC	
2	25	0	Inf. Dilution	lgK:-1.9(0.05SD)	0	ENS	5458
3	25	0	Inf. Dilution	lgK:1.9(1.SD)	0	CRV	6478

Note (3): Sign positive for rxn as written in JPD

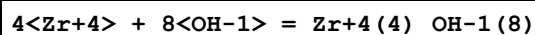
4	25	0	Inf. Dilution	lgK:7.000(1.600MD)	0	CRV	20829
5	25	0	Inf. Dilution	lgK:1.9(2SF)	0	EOD	29487
6	25	0	Inf. Dilution	lgK:7.54(3SF)	4	CRV	32222
7	25	0	Inf. Dilution	lgK:-1.86(3SF)	0	CRV	32224
8	90	0	Inf. Dilution	lgK:1.56(3SF)	0	EOD	29487
9	25	0	Inf. Dilution	dH:-10.845(5SF) kJ	0	EOD	29487

Reaction No. 55133



No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:1.0(2SF)	1	ELC	
2	25	0	Inf. Dilution	lgK:0.3(0.05SD)	0	ENS	5458
3	25	0	Inf. Dilution	lgK:0.320(0.220MD)	2	CRV	20829
4	25	0	Inf. Dilution	lgK:0.3(1SF)	3	CRV	23624
5	25	0	Inf. Dilution	lgK:0.27(0.13SD)	3	CRV	32222
6	25	0	Inf. Dilution	lgK:0.3(1SF)	0	CRV	32224
7	25	0.7	Seawater	lgK:-0.24(2SF)	0	ENS	6496
8	25	4	ClO4-1 salt	lgK:-0.6(0.2SD)	0	MIS	5077
9	20	4	HClO4	lgR:0.3(0.3SD)	3	MSE	5077
10	25	0	Inf. Dilution	dH:9.8(2SF)	3	CRV	6478
11	25	0.7	Seawater	dH:9.8(2SF)	0	ENS	6496

Reaction No. 55136



No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:118.5(4SF)	1	ELC	
2	25	0	Inf. Dilution	lgK:106.0(0.1SD)	0	CSM	8
Note (2): Low wgt for inter-reaction consistency							
3	25	2	Unknown	lgK:103.4(0.05SD)	0	CSM	8

Reaction No. 55137							
$3\text{<Zr+4>} + 4\text{<OH-1>} = \text{Zr+4(3)}_{\text{OH-1(4)}}$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:56.09(4SF)	1	ELC	
2	25	0	Inf. Dilution	lgK:55.4(3SF)	0	CRV	2556
3	25	2	Unknown	lgK:50.5(0.05SD)	0	CSM	8

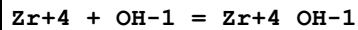
Reaction No. 55138							
$3\text{<Zr+4>} + 4\text{<H2O>} = \text{Zr+4(3)}_{\text{OH-1(4)}} + 4\text{<H+1>}$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:-0.6(0.05SD)	2	ENS	5458
2	25	0	Inf. Dilution	lgK:0.400(0.300MD)	3	CRV	20829
Note (2): Found not to be required by [18RKA_31367]							
3	25	0	Inf. Dilution	lgK:-0.6(1SF)	2	CRV	32224

Reaction No. 55139							
$\text{Zr+4} + 5\text{<H2O>} = \text{Zr+4}_{\text{OH-1(5)}} + 5\text{<H+1>}$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:-19.2(3SF)	1	ELC	
2	25	0	Inf. Dilution	lgK:-16.0(0.05SD)	0	ENS	5458
3	25	0	Inf. Dilution	lgK:-17.97(0.15SD)	4	CRV	32222
4	25	0	Inf. Dilution	lgK:-16.0(3SF)	0	CRV	32224
5	25	0.7	Seawater	lgK:-17.07(4SF)	0	ENS	6496
6	25	0.7	Seawater	dH:32.6(3SF)	2	ENS	6496

Reaction No. 55140							
$4\text{<Zr+4>} + 8\text{<H2O>} = \text{Zr+4(4)}_{\text{OH-1(8)}} + 8\text{<H+1>}$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:6.0(0.05SD)	2	ERS	5458
2	25	0	Inf. Dilution	lgK:6.520(0.650MD)	3	CRV	20829
Note (2): Found not to be required by [18RKA_31367]							

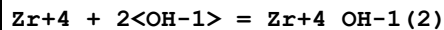
3	25	0	Inf. Dilution	lgK:6.0 (2SF)	2	CRV	32224
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Reaction No. 55351



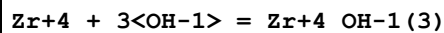
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:14.8 (3SF)	1	EOR	
2	25	0	Inf. Dilution	lgK:14.3 (3SF)	3	CRV	2556
3	25	0	Inf. Dilution	lgK:14.35 (0.09SD)	3	EOD	7033
4	25	0	Inf. Dilution	lgK:14.2 (3SF)	2	EOD	16981
5	25	0	Inf. Dilution	lgK:14.29 (4SF)	3	EOD	29848
6	25	0	Inf. Dilution	lgK:14.32 (0.22SD)	3	EOD	31367
7	25	0.1	HClO4	lgK:14.08 (4SF)	5	MDS	7033
8	25	0.3	Unknown	lgK:14.12 (4SF)	4	MDS	7033
9	25	0.5	Unknown	lgK:14.15 (4SF)	3	MDS	7033
10	25	1	(H,Na)ClO4	lgK:14.6 (3SF)	0	MSE	4444
11	25	1	(H,Na)ClO4	lgK:14.30 (4SF)	2	MDS	5076
12	25	1	ClO4-1 salt	lgK:14.6 (0.05SD)	0	MDS	4444
Note (12): Suspect other species							
13	25	1	ClO4-1 salt	lgK:14.32 (4SF)	2	MGL	5076
14	25	1	Unknown	lgK:14.30 (4SF)	2	MDS	7033
15	25	4	Unknown	lgK:13.9 (3SF)	5	CRV	2556

Reaction No. 55352



No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:29.8 (3SF)	1	ELC	
2	25	0	Inf. Dilution	lgK:28.26 (0.56SD)	0	EOD	7033
3	25	0	Inf. Dilution	lgK:27.1 (3SF)	0	EOD	16981
4	25	0	Inf. Dilution	lgK:26.66 (4SF)	0	EOD	29848
5	25	0	Inf. Dilution	lgK:25.7 (3SF)	0	CRV	31367
Note (5): <= 25.7							
6	25	1	ClO4-1 salt	lgK:28.8 (0.05SD)	0	MDS	4444
7	25	1	ClO4-1 salt	lgK:28.26 (4SF)	0	MGL	5076

Reaction No. 55353



No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
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1	25	0	Inf. Dilution	lgK:41.24(0.24SD)	4	EOD	7033
2	25	0	Inf. Dilution	lgK:35.85(4SF)	0	EOD	29848
3	25	1	ClO4-1 salt	lgK:42.7(0.08SD)	4	MDS	4444
4	25	1	ClO4-1 salt	lgK:41.92(4SF)	4	MGL	5076

Reaction No. 55354							
Zr+4 + 4<OH-1> = Zr+4_OH-1(4)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:53.81(4SF)	1	ELC	
2	25	0	Inf. Dilution	lgK:54.38(0.57SD)	2	EOD	7033
3	25	0	Inf. Dilution	lgK:43.12(4SF)	0	EOD	29848
4	25	0	Inf. Dilution	lgK:45.89(4SF)	0	CRV	31367
Note (4): < 45.89							
5	25	1	ClO4-1 salt	lgK:56.5(0.1SD)	2	MDS	4444
6	25	1	ClO4-1 salt	lgK:55.29(4SF)	2	MGL	5076

Reaction No. 55355							
Zr+4 + 3<NO3-1> = Zr+4_NO3-1(3)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	4	HNO3	lgR:-0.26(0.05SD)	5	MDS	5502

Reaction No. 55356							
Zr+4 + 4<NO3-1> = Zr+4_NO3-1(4)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:2.9(2SF)	1	ELC	
2	25	4	HNO3	lgR:-0.82(0.05SD)	5	MDS	5502

Reaction No. 55357							
Zr+4 + 4<Cl-1> = Zr+4_Cl-1(4)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	6.54	HCl	lgR:1.18(0.08SD)	5	MDS	5502

Reaction No. 55937							
Ca(s) + Zr(s) + 1.5<O2(g)> = CaO.ZrO2(s)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:285.0(4SF)	1	ELC	
2	25	0	Inf. Dilution	lgK:294.5(4SF)	0	ENS	6422

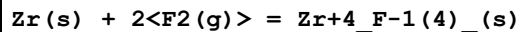
3	27	0	Inf. Dilution	lgK:292.6(4SF)	0	ENS	6422
4	127	0	Inf. Dilution	lgK:215.7(4SF)	0	ENS	6422
5	227	0	Inf. Dilution	lgK:169.6(4SF)	0	ENS	6422
6	327	0	Inf. Dilution	lgK:138.9(4SF)	0	ENS	6422
7	25	0	Inf. Dilution	dG:-401.8(0.05SD)	0	ENS	802
8	25	0	Inf. Dilution	dG:-401.8(4SF)	0	EFE	6422
9	25	0	Inf. Dilution	dH:-422.3(0.1SD)	2	ENS	802
10	25	0	Inf. Dilution	dH:-422.3(4SF)	2	MTD	6422
11	127	0	Inf. Dilution	dH:-422.1(4SF)	0	MTD	6422
12	227	0	Inf. Dilution	dH:-421.8(4SF)	0	MTD	6422
13	327	0	Inf. Dilution	dH:-421.4(4SF)	0	MTD	6422

Reaction No. 56354							
Ba(s) + Zr(s) + 1.5<O2(g)> = BaZrO3(s)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:287.4(4SF)	1	ELC	
2	25	0	Inf. Dilution	lgK:296.9(4SF)	0	ENS	6422
3	27	0	Inf. Dilution	lgK:295.0(4SF)	0	ENS	6422
4	127	0	Inf. Dilution	lgK:217.5(4SF)	0	ENS	6422
5	227	0	Inf. Dilution	lgK:171.1(4SF)	0	ENS	6422
6	327	0	Inf. Dilution	lgK:140.1(4SF)	0	ENS	6422
7	25	0	Inf. Dilution	dG:-405.0(0.05SD)	0	ENS	802
8	25	0	Inf. Dilution	dG:-405.0(4SF)	0	EFE	6422
9	25	0	Inf. Dilution	dH:-425.3(0.1SD)	0	ENS	802
10	25	0	Inf. Dilution	dH:-425.3(4SF)	0	MTD	6422
11	127	0	Inf. Dilution	dH:-425.1(4SF)	0	MTD	6422
12	227	0	Inf. Dilution	dH:-425.0(4SF)	0	MTD	6422
13	327	0	Inf. Dilution	dH:-425.0(4SF)	0	MTD	6422

Reaction No. 56395							
Sr(s) + Zr(s) + 1.5<O2(g)> = SrZrO3(s)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:294.6(4SF)	4	ENS	6422
2	27	0	Inf. Dilution	lgK:292.7(4SF)	0	ENS	6422
3	127	0	Inf. Dilution	lgK:215.8(4SF)	2	ENS	6422
4	227	0	Inf. Dilution	lgK:169.7(4SF)	1	ENS	6422

5	327	0	Inf. Dilution	lgK:139.0(4SF)	0	ENS	6422
6	25	0	Inf. Dilution	dG:-402.2(0.05SD)	4	ENS	802
7	25	0	Inf. Dilution	dG:-402.2(0.05SD)	5	ENS	802
8	25	0	Inf. Dilution	dG:-401.9(4SF)	4	EFE	6422
9	25	0	Inf. Dilution	dH:-422.4(0.1SD)	4	ENS	802
10	25	0	Inf. Dilution	dH:-422.4(0.1SD)	5	ENS	802
11	25	0	Inf. Dilution	dH:-422.4(4SF)	4	MTD	6422
12	127	0	Inf. Dilution	dH:-422.1(4SF)	1	MTD	6422
13	227	0	Inf. Dilution	dH:-421.7(4SF)	0	MTD	6422
14	327	0	Inf. Dilution	dH:-421.2(4SF)	0	MTD	6422

Reaction No. 56396



No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:317.1(4SF)	0	ENS	6422
2	27	0	Inf. Dilution	lgK:315.0(4SF)	0	ENS	6422
3	127	0	Inf. Dilution	lgK:231.9(4SF)	0	ENS	6422
4	227	0	Inf. Dilution	lgK:182.0(4SF)	0	ENS	6422
5	327	0	Inf. Dilution	lgK:148.9(4SF)	0	ENS	6422
6	25	0	Inf. Dilution	dG:-432.6(0.05SD)	0	ENS	802
7	25	0	Inf. Dilution	dG:-432.6(0.05SD)	0	ENS	802
8	25	0	Inf. Dilution	dG:-432.6(4SF)	0	EFE	6422
9	25	0	Inf. Dilution	dH:-456.8(0.1SD)	0	ENS	802
10	25	0	Inf. Dilution	dH:-456.8(0.1SD)	0	ENS	802
11	25	0	Inf. Dilution	dH:-456.8(4SF)	0	MTD	6422
12	127	0	Inf. Dilution	dH:-456.3(4SF)	0	MTD	6422
13	227	0	Inf. Dilution	dH:-455.8(4SF)	0	MTD	6422
14	327	0	Inf. Dilution	dH:-455.2(4SF)	0	MTD	6422

Reaction No. 56397



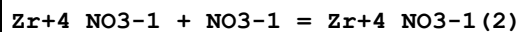
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:334.5(4SF)	2	ENS	6422
2	25	0	Inf. Dilution	lgK:336.3(4SF)	4	ENS	6494
3	27	0	Inf. Dilution	lgK:332.3(4SF)	0	ENS	6422
4	27	0	Inf. Dilution	lgK:334.1(4SF)	0	ENS	6494

5	127	0	Inf. Dilution	lgK:244.2(4SF)	2	ENS	6422
6	127	0	Inf. Dilution	lgK:245.6(4SF)	2	ENS	6494
7	227	0	Inf. Dilution	lgK:191.4(4SF)	1	ENS	6422
8	227	0	Inf. Dilution	lgK:192.5(4SF)	1	ENS	6494
9	327	0	Inf. Dilution	lgK:156.2(4SF)	0	ENS	6422
10	327	0	Inf. Dilution	lgK:157.1(4SF)	0	ENS	6494
11	25	0	Inf. Dilution	dG:-458.7(0.05SD)	4	ENS	802
12	25	0	Inf. Dilution	dG:-458.7(0.05SD)	5	ENS	802
13	25	0	Inf. Dilution	dG:-456.3(4SF)	0	EFE	6422
14	25	0	Inf. Dilution	dG:-458.8(4SF)	4	EFE	6494
15	25	0	Inf. Dilution	dG:-1919.7(3.1MD) kJ	6	CRV	20829
16	25	0	Inf. Dilution	dH:-486.0(0.1SD)	4	ENS	802
17	25	0	Inf. Dilution	dH:-486.0(0.1SD)	5	ENS	802
18	25	0	Inf. Dilution	dH:-483.7(4SF)	2	MTD	6422
19	25	0	Inf. Dilution	dH:-486.2(4SF)	4	MTD	6494
20	25	0	Inf. Dilution	dH:-2034.2(3.1MD) kJ	6	CRV	20829
21	127	0	Inf. Dilution	dH:-483.7(4SF)	1	MTD	6422
22	127	0	Inf. Dilution	dH:-486.1(4SF)	1	MTD	6494
23	227	0	Inf. Dilution	dH:-483.4(4SF)	0	MTD	6422
24	227	0	Inf. Dilution	dH:-485.9(4SF)	0	MTD	6494
25	327	0	Inf. Dilution	dH:-483.1(4SF)	0	MTD	6422
26	327	0	Inf. Dilution	dH:-485.5(4SF)	0	MTD	6494
27	25	0	Inf. Dilution	Cp:98.200(0.6MD) J/K	5	CRV	20829

Reaction No. 57698							
2<S(s)> + Zr(s) = Zr+4_S-2(2)_(s)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:99.86(4SF)	4	ENS	6422
2	27	0	Inf. Dilution	lgK:99.24(4SF)	0	ENS	6422
3	127	0	Inf. Dilution	lgK:74.08(4SF)	4	ENS	6422
4	227	0	Inf. Dilution	lgK:58.84(4SF)	4	ENS	6422
5	327	0	Inf. Dilution	lgK:48.62(4SF)	0	ENS	6422
6	25	0	Inf. Dilution	dG:-136.2(4SF)	4	EFE	6422
7	25	0	Inf. Dilution	dG:-565.750(12.0MD) kJ	6	CRV	20829
8	25	0	Inf. Dilution	dH:-138.0(4SF)	4	MTD	6422
9	25	0	Inf. Dilution	dH:-573.200(12.0MD) kJ	6	CRV	20829

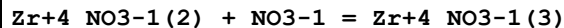
10	127	0	Inf. Dilution	dH:-139.1(4SF)	4	MTD	6422
11	227	0	Inf. Dilution	dH:-139.9(4SF)	4	MTD	6422
12	327	0	Inf. Dilution	dH:-140.6(4SF)	0	MTD	6422
13	25	0	Inf. Dilution	Cp:56.400(2.0MD)J/K	6	CRV	20829

Reaction No. 58378



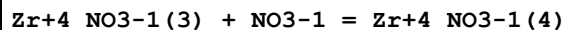
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	18:30	2	HClO4	lgK:-0.22(2SF)	4	MIX	22619
2	18:30	4	HClO4	lgK:-0.81(2SF)	4	MIX	22619
3	20	2	HClO4	lgK:-0.45(2SF)	4	MIX	13428
4	20	4	HClO4	lgK:-0.23(2SF)	0	MDS	140
5	23	2	HClO4	lgK:-0.30(2SF)	3	MIX	3262
6	23	4	HClO4	lgK:-0.80(2SF)	3	MIX	3262
7	25	0	Inf. Dilution	lgK:2.1(2SF)	1	EDH	
8	25	2	HClO4	lgK:-0.46(2SF)	5	MIX	13428
9	25	4	HClO4	lgK:0.11(2SF)	0	MIX	5502

Reaction No. 58379



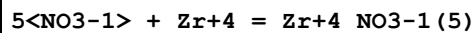
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	20	4	HClO4	lgK:-0.37(2SF)	3	MDS	140
2	25	4	HClO4	lgK:-0.26(2SF)	5	MIX	5502

Reaction No. 58380



No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	20	4	HClO4	lgK:-0.56(2SF)	3	MDS	140
2	25	4	HClO4	lgK:-0.82(2SF)	5	MIX	5502

Reaction No. 58381



No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	20	4	HClO4	lgK:-1.5(2SF)	3	MSC	140
2	25	0	Inf. Dilution	lgK:-0.803(3SF)	2	EAE	

Reaction No. 58382

6<NO3-1> + Zr+4 = Zr+4_NO3-1(6)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	20	4	HClO4	lgK:-1.7 (2SF)	3	MSC	140
2	25	0	Inf. Dilution	lgK:-1.07 (3SF)	2	EAE	

Reaction No. 58383							
2<NO3-1> + ZrO+2 = ZrO+2_NO3-1(2)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	2.5	HClO4	lgK:1.91 (3SF)	5	MDS	140

Reaction No. 58384							
4<NO3-1> + ZrO+2 = ZrO+2_NO3-1(4)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	2.5	HClO4	lgK:1.63 (3SF)	5	MDS	140

Reaction No. 58511							
Zr+4_SO4-2 + SO4-2 = Zr+4_SO4-2(2)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	20	2	HClO4	lgK:1.88 (3SF)	3	MIX	13428
2	20	2	HClO4	lgK:0.78 (2SF)	0	MSC	19258
3	20	4	HClO4	lgK:1.95 (3SF)	5	MSE	13458
4	23	2	HClO4	lgK:3.54 (3SF)	0	MIX	3262
5	25	0	Inf. Dilution	lgK:5.1 (2SF)	1	EES	
6	25	2	HClO4	lgK:2.80 (3SF)	0	MDS	140
7	25	2	HClO4	lgK:1.72 (3SF)	5	MSE	7998
8	25	2	HClO4	lgK:1.73 (3SF)	5	MSE	13652
9	25	2	NaClO4	lgK:2.85 (3SF)	0	MDS	8136
10	25	4	HClO4	lgK:1.85 (3SF)	5	MCE	22616
11	25	4	NaClO4	lgK:1.12 (3SF)	0	MDS	4708

Reaction No. 58512							
Zr+4_SO4-2(2) + SO4-2 = Zr+4_SO4-2(3)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	20	2	HClO4	lgK:1.00 (3SF)	3	MIX	13428
2	20	2	HClO4	lgK:2.27 (3SF)	0	MSC	19258
3	20	4	HClO4	lgK:0.30 (2SF)	5	MSE	13458
4	23	2	HClO4	lgK:6.59 (3SF)	0	MIX	3262

5	25	0	Inf. Dilution	lgK:1.7 (2SF)	1	EDH	
6	25	2	HClO4	lgK:1.1 (2SF)	3	MDS	140
7	25	2	HClO4	lgK:0.0 (0.3MD)	0	MSE	7998
8	25	2	HClO4	lgK:-0.08 (1SF)	0	MSE	13652

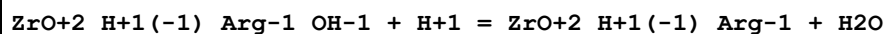
Reaction No. 58574							
Zr+4_Cl-1 + Cl-1 = Zr+4_Cl-1 (2)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	18:30	2	HClO4	lgK:-0.40 (2SF)	6	CRV	20829
2	18:30	2	HClO4	lgK:-0.90 (2SF)	4	MIX	22619
3	18:30	4	HClO4	lgK:-0.69 (2SF)	6	CRV	20829
4	18:30	4	HClO4	lgK:-0.72 (2SF)	4	MIX	22619
5	20	2	HClO4	lgK:-0.61 (2SF)	4	MIX	13428
6	20	6.54	H (ClO4)	lgK:0.40 (2SF)	3	MDS	140
7	23	2	HClO4	lgK:-0.90 (2SF)	3	MIX	3262
8	23	4	HClO4	lgK:-0.74 (2SF)	3	MIX	3262
9	25	0	Inf. Dilution	lgK:-0.65 (2SF)	2	EOD	7033
10	25	6.5	HClO4	lgK:1.32 (3SF)	3	MIX	5502

Reaction No. 58575							
Zr+4_Cl-1 (2) + Cl-1 = Zr+4_Cl-1 (3)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	18:30	2	HClO4	lgK:-0.23 (2SF)	4	MIX	22619
2	18:30	4	HClO4	lgK:-0.58 (2SF)	6	CRV	20829
3	18:30	4	HClO4	lgK:-0.62 (2SF)	4	MIX	22619
4	20	2	HClO4	lgK:-0.46 (2SF)	4	MIX	13428
5	20	6.54	H (ClO4)	lgK:0.19 (2SF)	3	MDS	140
6	23	2	HClO4	lgK:-0.23 (2SF)	3	MIX	3262
7	23	4	HClO4	lgK:-0.62 (2SF)	3	MIX	3262
8	25	0	Inf. Dilution	lgK:-1.08 (0.08SD)	2	EOD	7033

Reaction No. 58576							
Zr+4_Cl-1 (3) + Cl-1 = Zr+4_Cl-1 (4)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	18:30	2	HClO4	lgK:0.06 (1SF)	4	MIX	22619
2	18:30	4	HClO4	lgK:-0.30 (2SF)	6	CRV	20829
3	20	6.54	H (ClO4)	lgK:-0.33 (2SF)	3	MDS	140

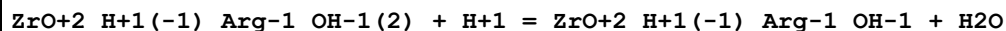
4	23	2	HClO4	lgK:0.06(1SF)	3	MIX	3262
5	25	0	Inf. Dilution	lgK:-1.10(3SF)	2	EOD	7033

Reaction No. 58713



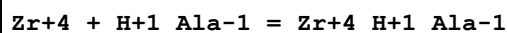
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0.1	KNO3	lgK:3.9(2SF)	0	MGL	140

Reaction No. 58714



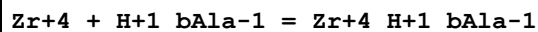
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0.1	KNO3	lgK:8.5(2SF)	0	MGL	140

Reaction No. 59997



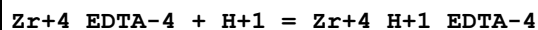
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	23	0.1	Unknown	lgK:4.1(0.1SD)	3	MSP	2392
2	25	0	Inf. Dilution	lgK:4.1(3SF)	2	EAE	

Reaction No. 60080



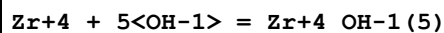
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	23	0.1	Unknown	lgK:5.3(0.1SD)	3	MSP	2392
2	25	0	Inf. Dilution	lgK:5.4(2SF)	1	EAE	

Reaction No. 60177



No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	20	1	NaClO4	lgK:6.1(2SF)	4	MRX	142
2	25	0	Inf. Dilution	lgK:6.1(3SF)	2	EAE	

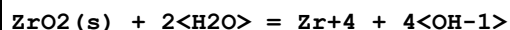
Reaction No. 60502



No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:52.0(3SF)	1	ELC	
2	25	0	Inf. Dilution	lgK:54.0(3SF)	0	CRV	2556
3	25	0	Inf. Dilution	lgK:46.52(4SF)	0	EOD	29848

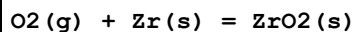
4	25	0	Inf. Dilution	lgK:50.35(0.23SD)	0	CRV	31367
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Reaction No. 60503



No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:-63.4(3SF)	1	ELC	
2	25	0	Inf. Dilution	lgK:-54.1(3SF)	0	CRV	2556
3	25	1	Unknown	lgK:-52.0(3SF)	0	CRV	2556

Reaction No. 60601



No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:182.2(4SF)	4	ENS	6422
2	25	0	Inf. Dilution	lgK:182.7(4SF)	4	ENS	6494
3	27	0	Inf. Dilution	lgK:181.0(4SF)	0	ENS	6422
4	27	0	Inf. Dilution	lgK:181.5(4SF)	0	ENS	6494
5	127	0	Inf. Dilution	lgK:133.2(4SF)	2	ENS	6422
6	127	0	Inf. Dilution	lgK:133.6(4SF)	2	ENS	6494
7	227	0	Inf. Dilution	lgK:104.6(4SF)	1	ENS	6422
8	227	0	Inf. Dilution	lgK:104.9(4SF)	1	ENS	6494
9	327	0	Inf. Dilution	lgK:85.49(4SF)	0	ENS	6422
10	327	0	Inf. Dilution	lgK:85.76(4SF)	0	ENS	6494
11	25	0	Inf. Dilution	dG:-248.5(4SF)	4	EFE	6422
12	25	0	Inf. Dilution	dG:-249.3(4SF)	4	EFE	6494
13	25	0	Inf. Dilution	dG:-1042.7(1.3MD) kJ	6	CRV	20829
14	25	0	Inf. Dilution	dG:-70.510(5SF)	0	CRV	31291
15	25	0	Inf. Dilution	dH:-262.3(4SF)	2	MTD	6422
16	25	0	Inf. Dilution	dH:-263.0(4SF)	2	MTD	6494
17	25	0	Inf. Dilution	dH:-1100.60(1.3MD) kJ	6	CRV	20829
18	127	0	Inf. Dilution	dH:-262.2(4SF)	1	MTD	6422
19	127	0	Inf. Dilution	dH:-262.9(4SF)	1	MTD	6494
20	227	0	Inf. Dilution	dH:-262.0(4SF)	0	MTD	6422
21	227	0	Inf. Dilution	dH:-262.7(4SF)	0	MTD	6494
22	327	0	Inf. Dilution	dH:-261.7(4SF)	0	MTD	6422
23	327	0	Inf. Dilution	dH:-262.5(4SF)	0	MTD	6494
24	25	0	Inf. Dilution	Cp:55.960(0.8MD) J/K	6	CRV	20829

Reaction No. 60630							
2<SO4-2> + Zr+4 = Zr+4_SO4-2 (2)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	20	4	Unknown	lgK:6.4 (2SF)	0	CRV	2556
2	25	0	Inf. Dilution	lgK:11.540 (0.210MD)	3	CRV	20829
3	25	0	Inf. Dilution	lgK:11.54 (4SF)	3	EOD	31756
4	25	2	NaClO4	lgK:6.54 (3SF)	0	MDS	7998
5	25	2	NaClO4	lgK:6.64 (3SF)	0	MDS	8136
6	25	2	Unknown	lgK:6.40 (3SF)	0	CRV	2556
7	25	4	NaClO4	lgK:2.72 (3SF)	0	MDS	4708

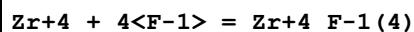
Reaction No. 60631							
3<SO4-2> + Zr+4 = Zr+4_SO4-2 (3)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	20	4	Unknown	lgK:7.5 (2SF)	0	CRV	2556
2	25	0	Inf. Dilution	lgK:14.3 (3SF)	1	ELC	
3	25	0	Inf. Dilution	lgK:14.300 (0.500MD)	0	CRV	20829
4	25	2	Unknown	lgK:7.4 (2SF)	0	CRV	2556

Reaction No. 60676							
2<F-1> + Zr+4 = Zr+4_F-1 (2)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:11.9 (3SF)	0	ELC	
2	25	0	Inf. Dilution	lgK:18.550 (0.310MD)	3	CRV	20829
3	25	2	Unknown	lgK:16.4 (3SF)	0	CRV	2556
4	25	4	Unknown	lgK:17.23 (4SF)	0	CRV	2556
5	25	0	Inf. Dilution	dH:-16.800 (1.00MD) kJ	3	CRV	20829
6	25	4	Unknown	dH:-2.5 (2SF)	0	CRV	2556

Reaction No. 60677							
3<F-1> + Zr+4 = Zr+4_F-1 (3)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:18.1 (3SF)	0	ELC	
2	25	0	Inf. Dilution	lgK:24.720 (0.380MD)	3	CRV	20829
3	25	2	Unknown	lgK:22.4 (3SF)	0	CRV	2556
4	25	4	Unknown	lgK:23.8 (3SF)	0	CRV	2556
5	25	0	Inf. Dilution	dH:-11.200 (1.70MD) kJ	3	CRV	20829

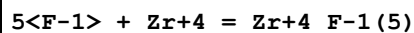
6	25	4	Unknown	dH:-2.3 (2SF)	0	CRV	2556
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Reaction No. 60678



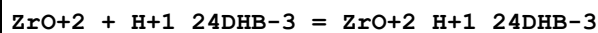
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:23.0 (3SF)	0	ELC	
2	25	0	Inf. Dilution	lgK:30.110 (0.400MD)	3	CRV	20829
3	25	4	Unknown	lgK:29.5 (3SF)	0	CRV	2556
4	25	0	Inf. Dilution	dH:-22.000 (2.700MD) kJ	3	CRV	20829
5	25	4	Unknown	dH:-4.7 (2SF)	0	CRV	2556

Reaction No. 60679



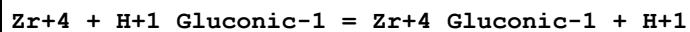
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	20	4	Unknown	lgK:34.6 (3SF)	0	CRV	2556
2	25	0	Inf. Dilution	lgK:27.3 (3SF)	0	ELC	
3	25	0	Inf. Dilution	lgK:34.600 (0.420MD)	3	CRV	20829

Reaction No. 60886



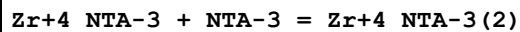
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	30	0.1	KNO3	lgK:16.55 (4SF)	6	MGL	5987

Reaction No. 61466



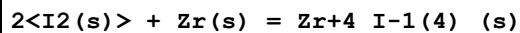
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	20	2	NaClO4	lgK:1.73 (3SF)	4	MIX	2261

Reaction No. 61487



No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	23			lgK:7.8 (0.2SD)	0	MSP	1118
2	25	0	Inf. Dilution	lgK:7.8 (2SF)	1	ESC	

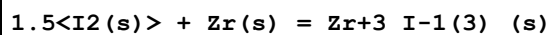
Reaction No. 66213



No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
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1	25	0	Inf. Dilution	lgK:85.05 (4SF)	4	ENS	6422
2	27	0	Inf. Dilution	lgK:84.52 (4SF)	0	ENS	6422
3	127	0	Inf. Dilution	lgK:63.09 (4SF)	4	ENS	6422
4	227	0	Inf. Dilution	lgK:48.60 (4SF)	4	ENS	6422
5	327	0	Inf. Dilution	lgK:38.06 (4SF)	0	ENS	6422
6	25	0	Inf. Dilution	dG:-116.0 (4SF)	4	EFE	6422
7	25	0	Inf. Dilution	dG:-480.860 (3.7MD) kJ	6	CRV	20829
8	25	0	Inf. Dilution	dG:-115.30 (5SF)	3	CRV	31291
9	25	0	Inf. Dilution	dH:-116.8 (4SF)	4	MTD	6422
10	25	0	Inf. Dilution	dH:-488.900 (3.7MD) kJ	6	CRV	20829
11	25	0	Inf. Dilution	dH:-116.80 (5SF)	3	CRV	31291
12	127	0	Inf. Dilution	dH:-124.7 (4SF)	1	MTD	6422
13	227	0	Inf. Dilution	dH:-145.2 (4SF)	0	MTD	6422
14	327	0	Inf. Dilution	dH:-144.4 (4SF)	0	MTD	6422
15	25	0	Inf. Dilution	Cp:125.100 (0.5MD) J/K	6	CRV	20829

Reaction No. 66214



No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:69.19 (4SF)	2	ENS	6422
2	27	0	Inf. Dilution	lgK:68.76 (4SF)	0	ENS	6422
3	127	0	Inf. Dilution	lgK:51.34 (4SF)	1	ENS	6422
4	227	0	Inf. Dilution	lgK:39.68 (4SF)	0	ENS	6422
5	327	0	Inf. Dilution	lgK:31.23 (4SF)	0	ENS	6422
6	25	0	Inf. Dilution	dG:-94.40 (4SF)	0	EFE	6422
7	25	0	Inf. Dilution	dG:-385.580 (15.0MD) kJ	0	CRV	20829
8	25	0	Inf. Dilution	dG:-92.986 (5SF)	2	CRV	31291
9	25	0	Inf. Dilution	dH:-95.0 (3SF)	0	MTD	6422
10	25	0	Inf. Dilution	dH:-391.600 (15.0MD) kJ	0	CRV	20829
11	25	0	Inf. Dilution	dH:-93.600 (5SF)	1	CRV	31291
12	127	0	Inf. Dilution	dH:-100.9 (4SF)	0	MTD	6422
13	227	0	Inf. Dilution	dH:-116.2 (4SF)	0	MTD	6422
14	327	0	Inf. Dilution	dH:-115.7 (4SF)	0	MTD	6422
15	25	0	Inf. Dilution	Cp:100.300 (0.2MD) J/K	0	CRV	20829

Reaction No. 66215

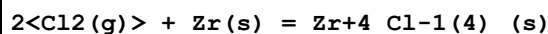
I2(s) + Zr(s) = Zr+2_I-1(2)_(s)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:45.20(4SF)	4	ENS	6422
2	27	0	Inf. Dilution	lgK:44.92(4SF)	0	ENS	6422
3	127	0	Inf. Dilution	lgK:33.58(4SF)	4	ENS	6422
4	227	0	Inf. Dilution	lgK:26.01(4SF)	0	ENS	6422
5	327	0	Inf. Dilution	lgK:20.54(4SF)	0	ENS	6422
6	25	0	Inf. Dilution	dG:-61.66(4SF)	4	EFE	6422
7	25	0	Inf. Dilution	dG:-273.800(10.0MD)kJ	0	CRV	20829
8	25	0	Inf. Dilution	dG:-66.049(5SF)	0	CRV	31291
9	25	0	Inf. Dilution	dH:-62.0(3SF)	1	MTD	6422
10	25	0	Inf. Dilution	dH:-277.800(10.0MD)kJ	3	CRV	20829
11	25	0	Inf. Dilution	dH:-66.400(5SF)	1	CRV	31291
12	127	0	Inf. Dilution	dH:-65.5(3SF)	3	MTD	6422
13	227	0	Inf. Dilution	dH:-75.4(3SF)	0	MTD	6422
14	327	0	Inf. Dilution	dH:-74.6(3SF)	0	MTD	6422
15	25	0	Inf. Dilution	Cp:75.500(0.2MD)J/K	6	CRV	20829

Reaction No. 66216							
Zr(s) + 1.5<F2(g)> = Zr+3_F-1(3)_(s)							
Note: Zirconium exists in aqueous solutions in the + 4 oxidation state only. See Brown et al., OECD Chem. Thermodynamics Ser., Vol. 8, 2005, p. 97							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:232.2(4SF)	0	ENS	6422
2	27	0	Inf. Dilution	lgK:230.7(4SF)	0	ENS	6422
3	127	0	Inf. Dilution	lgK:169.7(4SF)	0	ENS	6422
4	227	0	Inf. Dilution	lgK:133.2(4SF)	0	ENS	6422
5	327	0	Inf. Dilution	lgK:108.9(4SF)	0	ENS	6422
6	25	0	Inf. Dilution	dG:-316.8(4SF)	0	EFE	6422
7	25	0	Inf. Dilution	dH:-335.0(4SF)	0	MTD	6422
8	127	0	Inf. Dilution	dH:-334.6(4SF)	0	MTD	6422
9	227	0	Inf. Dilution	dH:-334.2(4SF)	0	MTD	6422
10	327	0	Inf. Dilution	dH:-333.7(4SF)	0	MTD	6422

Reaction No. 66217							
Zr(s) + F2(g) = Zr+2_F-1(2)_(s)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #

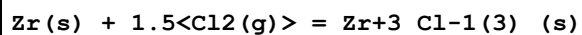
1	25	0	Inf. Dilution	lgK:159.9(4SF)	4	ENS	6422
2	27	0	Inf. Dilution	lgK:158.9(4SF)	0	ENS	6422
3	127	0	Inf. Dilution	lgK:117.0(4SF)	2	ENS	6422
4	227	0	Inf. Dilution	lgK:91.90(4SF)	1	ENS	6422
5	327	0	Inf. Dilution	lgK:75.20(4SF)	0	ENS	6422
6	25	0	Inf. Dilution	dG:-218.2(4SF)	4	EFE	6422
7	25	0	Inf. Dilution	dH:-230.0(4SF)	2	MTD	6422
8	127	0	Inf. Dilution	dH:-229.8(4SF)	1	MTD	6422
9	227	0	Inf. Dilution	dH:-229.5(4SF)	0	MTD	6422
10	327	0	Inf. Dilution	dH:-229.2(4SF)	0	MTD	6422

Reaction No. 66218



No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:155.8(4SF)	4	ENS	6422
2	27	0	Inf. Dilution	lgK:154.7(4SF)	0	ENS	6422
3	127	0	Inf. Dilution	lgK:112.1(4SF)	4	ENS	6422
4	227	0	Inf. Dilution	lgK:86.66(4SF)	3	ENS	6422
5	327	0	Inf. Dilution	lgK:69.73(4SF)	0	ENS	6422
6	25	0	Inf. Dilution	dG:-889.9(4SF) kJ	3	CRV	6330
7	25	0	Inf. Dilution	dG:-212.5(4SF)	4	EFE	6422
8	25	0	Inf. Dilution	dG:-890.150(1.2MD) kJ	4	CRV	20829
9	25	0	Inf. Dilution	dH:-980.5(4SF) kJ	3	CRV	6330
10	25	0	Inf. Dilution	dH:-234.2(4SF)	4	MTD	6422
11	25	0	Inf. Dilution	dH:-980.800(0.9MD) kJ	4	CRV	20829
12	127	0	Inf. Dilution	dH:-233.5(4SF)	4	MTD	6422
13	227	0	Inf. Dilution	dH:-232.8(4SF)	3	MTD	6422
14	327	0	Inf. Dilution	dH:-232.1(4SF)	0	MTD	6422
15	25	0	Inf. Dilution	Cp:117.500(1.6MD) J/K	6	CRV	20829

Reaction No. 66219



Note: Zirconium exists in aqueous solutions in the + 4 oxidation state only. See Brown et al., OECD Chem. Thermodynamics Ser., Vol. 8, 2005, p. 97

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:113.2(4SF)	0	ENS	6422
2	27	0	Inf. Dilution	lgK:112.4(4SF)	0	ENS	6422

3	127	0	Inf. Dilution	lgK:81.40 (4SF)	0	ENS	6422
4	227	0	Inf. Dilution	lgK:62.83 (4SF)	0	ENS	6422
5	327	0	Inf. Dilution	lgK:50.49 (4SF)	0	ENS	6422
6	25	0	Inf. Dilution	dG:-154.5 (4SF)	0	EFE	6422
7	25	0	Inf. Dilution	dG:-689.740 (3.0MD) kJ	0	CRV	20829
8	25	0	Inf. Dilution	dH:-170.7 (4SF)	0	MTD	6422
9	25	0	Inf. Dilution	dH:-760.100 (3.0MD) kJ	0	CRV	20829
10	127	0	Inf. Dilution	dH:-170.2 (4SF)	0	MTD	6422
11	227	0	Inf. Dilution	dH:-169.7 (4SF)	0	MTD	6422
12	327	0	Inf. Dilution	dH:-169.1 (4SF)	0	MTD	6422
13	25	0	Inf. Dilution	Cp:92.950 (0.6MD) J/K	0	CRV	20829

Reaction No. 66220							
Cl ₂ (g) + Zr(s) = Zr+2_Cl-1(2)_(s)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:67.56 (4SF)	0	ENS	6422
2	27	0	Inf. Dilution	lgK:67.10 (4SF)	0	ENS	6422
3	127	0	Inf. Dilution	lgK:48.37 (4SF)	0	ENS	6422
4	227	0	Inf. Dilution	lgK:37.17 (4SF)	0	ENS	6422
5	327	0	Inf. Dilution	lgK:29.73 (4SF)	0	ENS	6422
6	25	0	Inf. Dilution	dG:-92.17 (4SF)	0	EFE	6422
7	25	0	Inf. Dilution	dG:-496.150 (13.0MD) kJ	6	CRV	20829
8	25	0	Inf. Dilution	dH:-502.0 (4SF) kJ	0	CRV	6330
9	25	0	Inf. Dilution	dH:-103.0 (4SF)	0	MTD	6422
10	25	0	Inf. Dilution	dH:-542.800 (13.0MD) kJ	6	CRV	20829
11	127	0	Inf. Dilution	dH:-102.7 (4SF)	0	MTD	6422
12	227	0	Inf. Dilution	dH:-102.3 (4SF)	0	MTD	6422
13	327	0	Inf. Dilution	dH:-101.9 (4SF)	0	MTD	6422
14	25	0	Inf. Dilution	Cp:68.800 (1.6MD) J/K	6	CRV	20829

Reaction No. 66221							
Zr(s) + C(s) = ZrC(s)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:33.8 (3SF)	1	EDH	
2	25	0	Inf. Dilution	lgK:33.86 (4SF)	4	ENS	6422
3	27	0	Inf. Dilution	lgK:33.65 (4SF)	0	ENS	6422

4	127	0	Inf. Dilution	lgK:25.10 (4SF)	4	ENS	6422
5	227	0	Inf. Dilution	lgK:19.98 (4SF)	4	ENS	6422
6	327	0	Inf. Dilution	lgK:16.58 (4SF)	0	ENS	6422
7	25	0	Inf. Dilution	dG:-46.20 (4SF)	4	EFE	6422
8	25	0	Inf. Dilution	dG:-203.480 (2.6MD) kJ	0	CRV	20829
9	25	0	Inf. Dilution	dH:-47.0 (3SF)	0	MTD	6422
10	25	0	Inf. Dilution	dH:-207.000 (2.5MD) kJ	6	CRV	20829
11	127	0	Inf. Dilution	dH:-46.9 (3SF)	4	MTD	6422
12	227	0	Inf. Dilution	dH:-46.7 (3SF)	4	MTD	6422
13	327	0	Inf. Dilution	dH:-46.6 (3SF)	0	MTD	6422
14	25	0	Inf. Dilution	Cp:37.280 (0.7MD) J/K	6	CRV	20829

Reaction No. 66222							
2<Br2(l)> + Zr(s) = Zr+4_Br-1(4)_(s)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:127.1 (4SF)	3	ENS	6422
2	27	0	Inf. Dilution	lgK:126.2 (4SF)	0	ENS	6422
3	127	0	Inf. Dilution	lgK:91.53 (4SF)	2	ENS	6422
4	227	0	Inf. Dilution	lgK:70.17 (4SF)	3	ENS	6422
5	327	0	Inf. Dilution	lgK:55.97 (4SF)	0	ENS	6422
6	25	0	Inf. Dilution	dG:-173.3 (4SF)	4	EFE	6422
7	25	0	Inf. Dilution	dH:-760.7 (4SF) kJ	3	CRV	6330
8	25	0	Inf. Dilution	dH:-181.8 (4SF)	3	MTD	6422
9	25	0	Inf. Dilution	dH:-759.500 (3.3MD) kJ	5	CRV	20829
10	127	0	Inf. Dilution	dH:-195.9 (4SF)	0	MTD	6422
11	227	0	Inf. Dilution	dH:-195.1 (4SF)	0	MTD	6422
12	327	0	Inf. Dilution	dH:-194.4 (4SF)	0	MTD	6422

Reaction No. 66223							
Zr(s) + 1.5<Br2(l)> = Zr+3_Br-1(3)_(s)							
Note: Zirconium exists in aqueous solutions in the + 4 oxidation state only. See Brown et al., OECD Chem. Thermodynamics Ser., Vol. 8, 2005, p. 97							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:106.4 (4SF)	0	ENS	6422
2	27	0	Inf. Dilution	lgK:105.8 (4SF)	0	ENS	6422
3	127	0	Inf. Dilution	lgK:76.88 (4SF)	0	ENS	6422
4	227	0	Inf. Dilution	lgK:59.14 (4SF)	0	ENS	6422

5	327	0	Inf. Dilution	lgK:47.35(4SF)	0	ENS	6422
6	25	0	Inf. Dilution	dG:-145.2(4SF)	0	EFE	6422
7	25	0	Inf. Dilution	dH:-152.0(4SF)	0	MTD	6422
8	25	0	Inf. Dilution	dH:-599.800(13.0MD)kJ	0	CRV	20829
9	127	0	Inf. Dilution	dH:-162.6(4SF)	0	MTD	6422
10	227	0	Inf. Dilution	dH:-162.1(4SF)	0	MTD	6422
11	327	0	Inf. Dilution	dH:-161.5(4SF)	0	MTD	6422

Reaction No. 66224							
Zr(s) + Br2(l) = Zr+2_Br-1(2)_(s)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:66.96(4SF)	4	ENS	6422
2	27	0	Inf. Dilution	lgK:66.52(4SF)	0	ENS	6422
3	127	0	Inf. Dilution	lgK:48.13(4SF)	2	ENS	6422
4	227	0	Inf. Dilution	lgK:36.86(4SF)	1	ENS	6422
5	327	0	Inf. Dilution	lgK:29.38(4SF)	0	ENS	6422
6	25	0	Inf. Dilution	dG:-91.35(4SF)	4	EFE	6422
7	25	0	Inf. Dilution	dH:-96.7(3SF)	4	MTD	6422
8	25	0	Inf. Dilution	dH:-399.200(13.0MD)kJ	0	CRV	20829
9	127	0	Inf. Dilution	dH:-103.5(4SF)	1	MTD	6422
10	227	0	Inf. Dilution	dH:-102.9(4SF)	0	MTD	6422
11	327	0	Inf. Dilution	dH:-102.3(4SF)	0	MTD	6422

Reaction No. 66264							
2<Y(s)> + 2<Zr(s)> + 3.5<O2(g)> = Y2Zr2O7(s)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:686.4(4SF)	4	ENS	6422
2	27	0	Inf. Dilution	lgK:681.9(4SF)	0	ENS	6422
3	127	0	Inf. Dilution	lgK:502.6(4SF)	2	ENS	6422
4	227	0	Inf. Dilution	lgK:395.0(4SF)	1	ENS	6422
5	327	0	Inf. Dilution	lgK:323.3(4SF)	0	ENS	6422
6	25	0	Inf. Dilution	dG:-936.4(4SF)	4	EFE	6422
7	25	0	Inf. Dilution	dH:-985.2(4SF)	2	MTD	6422
8	127	0	Inf. Dilution	dH:-984.7(4SF)	1	MTD	6422
9	227	0	Inf. Dilution	dH:-983.9(4SF)	0	MTD	6422
10	327	0	Inf. Dilution	dH:-983.0(4SF)	0	MTD	6422

Reaction No. 66418							
$2\text{<Sm(s)>} + 2\text{<Zr(s)>} + 3.5\text{<O}_2\text{(g)>} = \text{Sm}_2\text{Zr}_2\text{O}_7\text{(s)}$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:677.6(4SF)	4	ENS	6422
2	27	0	Inf. Dilution	lgK:673.2(4SF)	0	ENS	6422
3	127	0	Inf. Dilution	lgK:496.0(4SF)	2	ENS	6422
4	227	0	Inf. Dilution	lgK:389.8(4SF)	1	ENS	6422
5	327	0	Inf. Dilution	lgK:319.0(4SF)	0	ENS	6422
6	25	0	Inf. Dilution	dG:-924.3(4SF)	4	EFE	6422
7	25	0	Inf. Dilution	dH:-973.0(4SF)	2	MTD	6422
8	127	0	Inf. Dilution	dH:-972.4(4SF)	1	MTD	6422
9	227	0	Inf. Dilution	dH:-971.7(4SF)	0	MTD	6422
10	327	0	Inf. Dilution	dH:-971.0(4SF)	0	MTD	6422

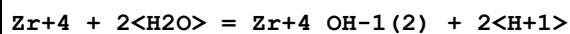
Reaction No. 66842							
$\text{ZrO}+2\text{_{H+1(4)}_{PO4-3(2)}}\text{(s)} + \text{H}_2\text{O} = \text{Zr}+4 + 2\text{<OH-1>} + 2\text{<H+1(2)}_{\text{PO4-3}}$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	23	0	Unknown	lgK:-17.64(4SF)	4	ENS	773
2	25	0	Inf. Dilution	lgK:-17.6(3SF)	2	EAE	

Reaction No. 66959							
$2\text{<Li(s)>} + \text{Zr(s)} + 1.5\text{<O}_2\text{(g)>} = \text{Li}_2\text{ZrO}_3\text{(s)}$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:292.0(4SF)	4	ENS	6422
2	27	0	Inf. Dilution	lgK:290.1(4SF)	0	ENS	6422
3	127	0	Inf. Dilution	lgK:213.5(4SF)	2	ENS	6422
4	227	0	Inf. Dilution	lgK:167.5(4SF)	1	ENS	6422
5	327	0	Inf. Dilution	lgK:136.7(4SF)	0	ENS	6422
6	25	0	Inf. Dilution	dG:-398.4(4SF)	4	EFE	6422
7	25	0	Inf. Dilution	dH:-420.7(4SF)	2	MTD	6422
8	127	0	Inf. Dilution	dH:-420.6(4SF)	1	MTD	6422
9	227	0	Inf. Dilution	dH:-422.0(4SF)	0	MTD	6422
10	327	0	Inf. Dilution	dH:-421.9(4SF)	0	MTD	6422

Reaction No. 67090							
$2\text{<Nd(s)>} + 3.5\text{<O}_2\text{(g)>} + 2\text{<Zr(s)>} = \text{Nd}_2\text{O}_3.2\text{ZrO}_2\text{(s)}$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #

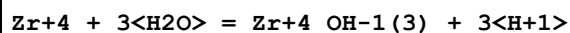
1	25	0	Inf. Dilution	lgK:673.6(4SF)	4	ENS	6422
2	27	0	Inf. Dilution	lgK:669.2(4SF)	0	ENS	6422
3	127	0	Inf. Dilution	lgK:493.1(4SF)	2	ENS	6422
4	227	0	Inf. Dilution	lgK:387.5(4SF)	1	ENS	6422
5	327	0	Inf. Dilution	lgK:317.2(4SF)	0	ENS	6422
6	25	0	Inf. Dilution	dG:-918.9(4SF)	4	EFE	6422
7	25	0	Inf. Dilution	dH:-967.2(4SF)	2	MTD	6422
8	127	0	Inf. Dilution	dH:-966.6(4SF)	1	MTD	6422
9	227	0	Inf. Dilution	dH:-965.7(4SF)	0	MTD	6422
10	327	0	Inf. Dilution	dH:-964.8(4SF)	0	MTD	6422

Reaction No. 67451



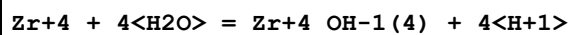
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:1.9(2SF)	1	ELC	
2	25	0	Inf. Dilution	lgK:0.980(1.060MD)	0	CRV	20829
3	25	0	Inf. Dilution	lgK:2.7(2SF)	0	CRV	32222
Note (3): < 2.7 (seems odd)							
4	25	0	Inf. Dilution	lgK:-1.7(2SF)	0	CRV	32224
5	25	0.7	Seawater	lgK:-2.79(3SF)	0	ENS	6496
Note (5): p. 984							
6	25	0.7	Seawater	dH:18.1(3SF)	0	ENS	6496
Note (6): p .984							

Reaction No. 67452



No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:-0.8(1SF)	1	ELC	
2	25	0	Inf. Dilution	lgK:-4.5(2SF)	0	CRV	32222
Note (2): < -4.5							
3	25	0	Inf. Dilution	lgK:-5.1(2SF)	0	CRV	32224
4	25	0.7	Seawater	lgK:-6.48(3SF)	0	ENS	6496
5	25	0.7	Seawater	dH:24.7(3SF)	0	ENS	6496

Reaction No. 67453



No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
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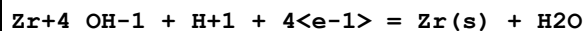
1	25	0	Inf. Dilution	lgK:-1.7 (2SF)	1	ELC	
2	25	0	Inf. Dilution	lgK:-2.190 (1.700MD)	0	CRV	20829
3	25	0	Inf. Dilution	lgK:-5.13 (0.36SD)	0	EOD	32222
4	25	0	Inf. Dilution	lgK:-9.7 (2SF)	0	CRV	32224
5	25	0.7	Seawater	lgK:-11.08 (4SF)	0	ENS	6496
6	25	0.7	Seawater	dH:29.3 (3SF)	0	ENS	6496

Reaction No. 70917



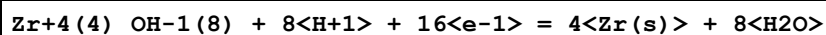
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:-99.4 (3SF)	1	ELC	
2	25	0	Inf. Dilution	E:-1.553 (4SF)	0	CRV	6330
3	25	0	Inf. Dilution	E:-1.473 (4SF)	0	CRV	7761
4	25	0	Inf. Dilution	E:-1.553 (4SF)	0	CRV	22519
5	25	0	Inf. Dilution	E:-1.45 (3SF)	0	CRV	22551
6	25	0	Inf. Dilution	dH:528.9 (4SF) kJ	0	CRV	7761

Reaction No. 71654



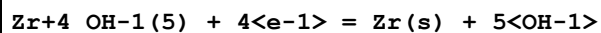
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:-93.0 (3SF)	1	ELC	
2	25	0	Inf. Dilution	E:-1.45 (3SF)	0	CRV	7761

Reaction No. 71655



No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:-378.7 (4SF)	0	ELC	
Note (1): from DOCMN using only SSS							
2	25	0	Inf. Dilution	lgK:-374.6 (4SF)	1	ELC	
3	25	0	Inf. Dilution	E:-1.47 (3SF)	0	CRV	7761

Reaction No. 71656



No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:-144.0 (4SF)	1	ELC	
2	25	0	Inf. Dilution	E:-2.22 (3SF)	0	CRV	7761

Reaction No. 71657							
$\text{ZrO}_2(\text{am.}, \text{s}) + 2\langle\text{H}_2\text{O}\rangle + 4\langle\text{e}^{-1}\rangle = \text{Zr}(\text{s}) + 4\langle\text{OH}^{-1}\rangle$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	$\lg K: -148.2 (4\text{SF})$	1	ELC	
2	25	0	Inf. Dilution	$E: -2.28 (3\text{SF})$	0	CRV	7761
3	25	0	Inf. Dilution	$\text{dH}: 743.0 (4\text{SF}) \text{ kJ}$	2	CRV	7761

Reaction No. 71658							
$\text{ZrO}_2(\text{s}) + 2\langle\text{H}_2\text{O}\rangle + 4\langle\text{e}^{-1}\rangle = \text{Zr}(\text{s}) + 4\langle\text{OH}^{-1}\rangle$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	$\lg K: -155.4 (4\text{SF})$	1	ELC	
2	25	0	Inf. Dilution	$E: -2.301 (4\text{SF})$	0	CRV	7761
3	25	0	Inf. Dilution	$\text{dH}: 752.3 (4\text{SF}) \text{ kJ}$	2	CRV	7761

Reaction No. 73371							
$\text{Zr+4_OH-1} + \text{OH-1} = \text{Zr+4_OH-1 (2)}$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	$\lg K: 14.9 (3\text{SF})$	2	EDH	
2	25	1	(H,Na)ClO ₄	$\lg K: 14.2 (3\text{SF})$	3	MSE	4444
3	25	1	ClO ₄ -1 salt	$\lg K: 13.94 (4\text{SF})$	4	MGL	5076

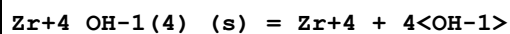
Reaction No. 73372							
$\text{Zr+4_OH-1 (2)} + \text{OH-1} = \text{Zr+4_OH-1 (3)}$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	$\lg K: 11.3 (3\text{SF})$	1	ELC	
2	25	1	(H,Na)ClO ₄	$\lg K: 13.9 (3\text{SF})$	0	MSE	4444
3	25	1	ClO ₄ -1 salt	$\lg K: 13.65 (4\text{SF})$	0	MGL	5076

Reaction No. 73373							
$\text{Zr+4_OH-1 (3)} + \text{OH-1} = \text{Zr+4_OH-1 (4)}$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	$\lg K: 12.6 (3\text{SF})$	1	ELC	
2	25	1	(H,Na)ClO ₄	$\lg K: 13.8 (3\text{SF})$	0	MSE	4444
3	25	1	ClO ₄ -1 salt	$\lg K: 13.36 (4\text{SF})$	0	MGL	5076

Reaction No. 73876							
$\text{ZrO}(\text{OH})_2(\text{s}) + \text{H}_2\text{O} + 4\langle\text{e}^{-1}\rangle = \text{Zr}(\text{s}) + 4\langle\text{OH}^{-1}\rangle$							

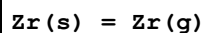
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	E:-2.36(3SF)	5	CRV	22519

Reaction No. 74031



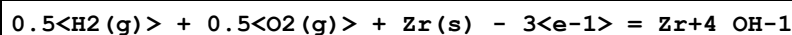
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:-59.24(4SF)	1	ELC	
2	25	0	Inf. Dilution	lgK:-52.0(3SF)	0	EOD	16981
3	25	0	Inf. Dilution	lgK:-56.(2SF)	0	CRV	22551
4	25	0	Inf. Dilution	lgK:-56.94(4SF)	0	EOD	29848

Reaction No. 74536



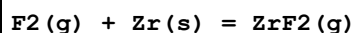
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	dG:566.880(1.4MD)kJ	6	CRV	20829
2	25	0	Inf. Dilution	dH:609.800(1.4MD)kJ	6	CRV	20829
3	25	0	Inf. Dilution	Cp:24.775(0.01MD)J/K	6	CRV	20829

Reaction No. 74537



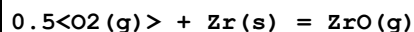
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:134.6(4SF)	1	ELC	
2	25	0	Inf. Dilution	dG:-767.480(9.312MD)kJ	0	CRV	20829

Reaction No. 74538



No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	dG:-492.770(1.436MD)kJ	6	CRV	20829
2	25	0	Inf. Dilution	dH:-478.300(0.800MD)kJ	6	CRV	20829
3	25	0	Inf. Dilution	Cp:49.400(2.000MD)J/K	6	CRV	20829

Reaction No. 74539



No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	dG:30.576(27.1MD)kJ	6	CRV	20829
2	25	0	Inf. Dilution	dH:56.200(27.0MD)kJ	6	CRV	20829
3	25	0	Inf. Dilution	Cp:30.800(1.0MD)J/K	6	CRV	20829

Reaction No. 74540							
$O_2(g) + Zr(s) = ZrO_2(g)$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	dG:-298.390(47.0MD)kJ	6	CRV	20829
2	25	0	Inf. Dilution	dH:-289.600(47.0MD)kJ	6	CRV	20829
3	25	0	Inf. Dilution	Cp:46.100(1.0MD)J/K	6	CRV	20829

Reaction No. 74541							
$0.5<H_2(g)> + Zr(s) = ZrH(s)$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	dG:-64.686(0.8MD)kJ	6	CRV	20829
2	25	0	Inf. Dilution	dH:-84.400(0.8MD)kJ	6	CRV	20829
3	25	0	Inf. Dilution	Cp:30.200(0.1MD)J/K	6	CRV	20829

Reaction No. 74542							
$H_2(g) + Zr(s) = ZrH_2(s)$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	dG:-124.280(1.7MD)kJ	6	CRV	20829
2	25	0	Inf. Dilution	dH:-164.400(1.7MD)kJ	6	CRV	20829
3	25	0	Inf. Dilution	Cp:30.930(0.06MD)J/K	6	CRV	20829

Reaction No. 74543							
$H_2(g) + O_2(g) + Zr(s) - 2<e-1> = Zr+4_OH-1(2)$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:177.0(4SF)	1	ELC	
2	25	0	Inf. Dilution	dG:-1008.4(11.0MD)kJ	0	CRV	20829

Reaction No. 74544							
$2<H_2(g)> + 2<O_2(g)> + Zr(s) = Zr+4_OH-1(4)$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:256.0(4SF)	1	ELC	
2	25	0	Inf. Dilution	dG:-1464.6(13.3MD)kJ	0	CRV	20829

Reaction No. 74545							
$3<H_2(g)> + 3<O_2(g)> + Zr(s) + 2<e-1> = Zr+4_OH-1(6)$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #

1	25	0	Inf. Dilution	lgK:308.2(4SF)	1	ELC	
2	25	0	Inf. Dilution	dG:-1785.8(10.1MD)kJ	0	CRV	20829

Reaction No. 74546							
$2\text{H}_2(\text{g}) + 2\text{O}_2(\text{g}) + 3\text{Zr}(\text{s}) - 8\text{e}^- = \text{Zr}_3\text{O}_7(\text{s})$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:442.3(4SF)	1	ELC	
2	25	0	Inf. Dilution	dG:-2536.4(27.7MD)kJ	0	CRV	20829
3	25	0	Inf. Dilution	dH:-2970.8(10.0MD)kJ	2	CRV	20829
4	25	0	Inf. Dilution	Cp:-1190.9(98.8MD)J/K	2	CRV	20829

Reaction No. 74547							
$4.5\text{H}_2(\text{g}) + 4.5\text{O}_2(\text{g}) + 3\text{Zr}(\text{s}) - 3\text{e}^- = \text{Zr}_3\text{O}_7(\text{s})$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:665.3(4SF)	0	ELC	
Note (1): from DOCMN using only SSS							
2	25	0	Inf. Dilution	lgK:662.3(4SF)	1	ELC	
3	25	0	Inf. Dilution	dG:-3789.4(27.7MD)kJ	0	CRV	20829

Reaction No. 74548							
$4\text{H}_2(\text{g}) + 4\text{O}_2(\text{g}) + 4\text{Zr}(\text{s}) - 8\text{e}^- = \text{Zr}_4\text{O}_{10}(\text{s})$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:711.1(4SF)	0	ELC	
Note (1): from DOCMN using only SSS							
2	25	0	Inf. Dilution	lgK:707.0(4SF)	1	ELC	
3	25	0	Inf. Dilution	dG:-4048.4(37.1MD)kJ	0	CRV	20829

Reaction No. 74549							
$7.5\text{H}_2(\text{g}) + 7.5\text{O}_2(\text{g}) + 4\text{Zr}(\text{s}) - \text{e}^- = \text{Zr}_4\text{O}_{10}(\text{s})$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:1008.0(5SF)	0	ELC	
Note (1): from DOCMN using only SSS							
2	25	0	Inf. Dilution	lgK:1004.(4SF)	1	ELC	
3	25	0	Inf. Dilution	dG:-5743.0(36.9MD)kJ	0	CRV	20829

Reaction No. 74550							
$8\text{H}_2(\text{g}) + 8\text{O}_2(\text{g}) + 4\text{Zr}(\text{s}) = \text{Zr}_4\text{O}_{10}(\text{s})$							

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:1045.0(5SF)	0	ELC	
Note (1): from DOCMN using only SSS							
2	25	0	Inf. Dilution	lgK:1041.(4SF)	1	ELC	
3	25	0	Inf. Dilution	dG:-5956.2(37.2MD)kJ	0	CRV	20829
4	25	0	Inf. Dilution	dH:-6706.20(7.2MD)kJ	0	CRV	20829

Reaction No. 74551							
$0.5\text{<F2(g)>} + \text{Zr(s)} = \text{ZrF(g)}$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	dG:41.353(1.2MD)kJ	6	CRV	20829
2	25	0	Inf. Dilution	dH:72.100(1.0MD)kJ	6	CRV	20829
3	25	0	Inf. Dilution	Cp:32.800(1.0MD)J/K	6	CRV	20829

Reaction No. 74552							
$0.5\text{<F2(g)>} + \text{Zr(s)} - 3\text{<e-1>} = \text{Zr+4_F-1}$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:148.3(4SF)	0	ELC	
2	25	0	Inf. Dilution	dG:-867.800(9.2MD)kJ	3	CRV	20829
3	25	0	Inf. Dilution	dH:-961.350(5.1MD)kJ	3	CRV	20829

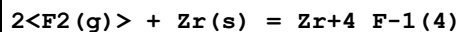
Reaction No. 74553							
$\text{F2(g)} + \text{Zr(s)} - 2\text{<e-1>} = \text{Zr+4_F-1(2)}$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:208.9(4SF)	1	ELC	
2	25	0	Inf. Dilution	dG:-1197.4(9.5MD)kJ	0	CRV	20829
3	25	0	Inf. Dilution	dH:-1296.0(5.2MD)kJ	2	CRV	20829

Reaction No. 74554							
$1.5\text{<F2(g)>} + \text{Zr(s)} = \text{ZrF3(g)}$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	dG:-1065.7(1.7MD)kJ	6	CRV	20829
2	25	0	Inf. Dilution	dH:-1075.9(1.2MD)kJ	6	CRV	20829
3	25	0	Inf. Dilution	Cp:68.400(2.0MD)J/K	6	CRV	20829

Reaction No. 74555							
$1.5\text{<F2(g)>} + \text{Zr(s)} - \text{e-1} = \text{Zr+4_F-1(3)}$							

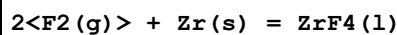
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:258.6(4SF)	0	ELC	
2	25	0	Inf. Dilution	dG:-1514.2(9.7MD)kJ	0	CRV	20829
3	25	0	Inf. Dilution	dH:-1625.8(5.6MD)kJ	2	CRV	20829

Reaction No. 74556



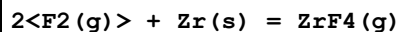
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:318.8(4SF)	1	ELC	
2	25	0	Inf. Dilution	dG:-1826.5(9.9MD)kJ	0	CRV	20829
3	25	0	Inf. Dilution	dH:-1971.9(6.2MD)kJ	2	CRV	20829

Reaction No. 74557



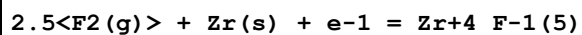
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	dH:-1975.5(1.1MD)kJ	6	CRV	20829

Reaction No. 74558



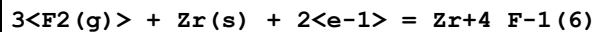
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	dG:-1634.6(1.0MD)kJ	6	CRV	20829
2	25	0	Inf. Dilution	dH:-1671.1(1.0MD)kJ	6	CRV	20829
3	25	0	Inf. Dilution	Cp:87.440(0.5MD)J/K	6	CRV	20829

Reaction No. 74559



No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:372.5(4SF)	1	ELC	
2	25	0	Inf. Dilution	dG:-2133.5(10.0MD)kJ	0	CRV	20829

Reaction No. 74560



No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:425.1(4SF)	1	ELC	
2	25	0	Inf. Dilution	dG:-2435.1(10.3MD)kJ	0	CRV	20829

Reaction No. 74561

0.5<Cl ₂ (g)> + Zr(s) = ZrCl(s)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	dG:-264.480 (2.0MD) kJ	6	CRV	20829
2	25	0	Inf. Dilution	dH:-291.200 (2.0MD) kJ	6	CRV	20829
3	25	0	Inf. Dilution	Cp:47.160 (0.6MD) J/K	6	CRV	20829

Reaction No. 74562							
0.5<Cl ₂ (g)> + Zr(s) = ZrCl(g)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	dG:200.420 (23.6MD) kJ	6	CRV	20829
2	25	0	Inf. Dilution	dH:231.840 (23.6MD) kJ	6	CRV	20829
3	25	0	Inf. Dilution	Cp:35.700 (1.5MD) J/K	6	CRV	20829

Reaction No. 74563							
0.5<Cl ₂ (g)> + Zr(s) - 3<e-1> = Zr+4_Cl-1							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:116.1 (4SF)	1	ELC	
2	25	0	Inf. Dilution	dG:-668.800 (9.2MD) kJ	0	CRV	20829

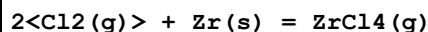
Reaction No. 74564							
Cl ₂ (g) + Zr(s) = ZrCl ₂ (g)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	dG:-154.310 (19.9MD) kJ	6	CRV	20829
2	25	0	Inf. Dilution	dH:-145.390 (19.8MD) kJ	6	CRV	20829
3	25	0	Inf. Dilution	Cp:58.300 (2.5MD) J/K	6	CRV	20829

Reaction No. 74565							
Cl ₂ (g) + Zr(s) - 2<e-1> = Zr+4_Cl-1(2)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:138.5 (4SF)	1	ELC	
2	25	0	Inf. Dilution	dG:-803.330 (9.3MD) kJ	0	CRV	20829

Reaction No. 74566							
1.5<Cl ₂ (g)> + Zr(s) = ZrCl ₃ (g)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	dG:-510.570 (9.2MD) kJ	6	CRV	20829
2	25	0	Inf. Dilution	dH:-520.140 (9.1MD) kJ	6	CRV	20829

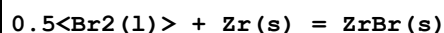
3	25	0	Inf. Dilution	Cp:76.100 (2.5MD) J/K	6	CRV	20829
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Reaction No. 74567



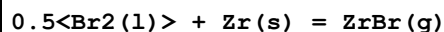
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	dG:-836.430 (0.9MD) kJ	6	CRV	20829
2	25	0	Inf. Dilution	dH:-871.500 (0.9MD) kJ	6	CRV	20829
3	25	0	Inf. Dilution	Cp:98.200 (1.6MD) J/K	6	CRV	20829

Reaction No. 74568



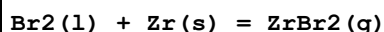
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	dG:-180. (3SF) kJ	1	EES	
2	25	0	Inf. Dilution	dH:-218.90 (3.5MD) kJ	6	CRV	20829

Reaction No. 74569



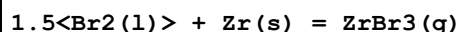
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	Cp:36.700 (1.5MD) J/K	6	CRV	20829

Reaction No. 74570



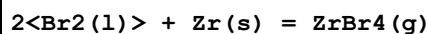
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	Cp:60.400 (2.5MD) J/K	6	CRV	20829

Reaction No. 74571



No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	Cp:79.400 (2.5MD) J/K	6	CRV	20829

Reaction No. 74572



No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	dG:-664.670 (3.8MD) kJ	6	CRV	20829
2	25	0	Inf. Dilution	dH:-643.500 (3.7MD) kJ	6	CRV	20829
3	25	0	Inf. Dilution	Cp:102.700 (1.0MD) J/K	6	CRV	20829

Reaction No. 74573

0.5<I2(s)> + Zr(s) = ZrI(s)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	dG:-150.420 (1.2MD) kJ	6	CRV	20829
2	25	0	Inf. Dilution	dH:-152.400 (1.2MD) kJ	6	CRV	20829
3	25	0	Inf. Dilution	Cp:50.800 (0.1MD) J/K	6	CRV	20829

Reaction No. 74574							
0.5<I2(s)> + Zr(s) = ZrI(g)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	dG:349.320 (11.0MD) kJ	6	CRV	20829
2	25	0	Inf. Dilution	dH:403.000 (11.0MD) kJ	6	CRV	20829
3	25	0	Inf. Dilution	Cp:37.200 (1.5MD) J/K	6	CRV	20829

Reaction No. 74575							
I2(s) + Zr(s) = ZrI2(g)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	dG:77.761 (16.0MD) kJ	6	CRV	20829
2	25	0	Inf. Dilution	dH:135.000 (16.0MD) kJ	6	CRV	20829
3	25	0	Inf. Dilution	Cp:61.400 (2.5MD) J/K	6	CRV	20829

Reaction No. 74576							
1.5<I2(s)> + Zr(s) = ZrI3(g)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	dG:-185.880 (4.4MD) kJ	6	CRV	20829
2	25	0	Inf. Dilution	dH:-130.000 (4.1MD) kJ	6	CRV	20829
3	25	0	Inf. Dilution	Cp:80.900 (2.5MD) J/K	6	CRV	20829

Reaction No. 74577							
2<I2(s)> + Zr(s) = ZrI4(g)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	dG:-405.960 (3.7MD) kJ	6	CRV	20829
2	25	0	Inf. Dilution	dH:-355.500 (3.7MD) kJ	6	CRV	20829
3	25	0	Inf. Dilution	Cp:104.590 (0.2MD) J/K	6	CRV	20829

Reaction No. 74578							
3<S(s)> + Zr(s) = ZrS3(s)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #

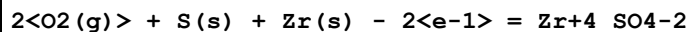
1	25	0	Inf. Dilution	dG:-601.920 (4.8MD) kJ	6	CRV	20829
2	25	0	Inf. Dilution	dH:-615.200 (4.6MD) kJ	6	CRV	20829

Reaction No. 74579



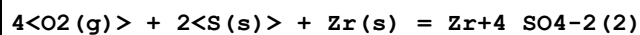
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	dG:-1834.9 (10.0MD) kJ	6	CRV	20829
2	25	0	Inf. Dilution	dH:-2011.0 (10.0MD) kJ	6	CRV	20829

Reaction No. 74580



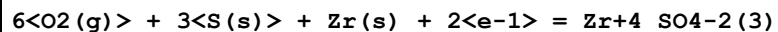
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:229.0 (4SF)	1	ELC	
2	25	0	Inf. Dilution	dG:-1312.7 (9.2MD) kJ	0	CRV	20829
3	25	0	Inf. Dilution	dH:-1480.9 (5.1MD) kJ	3	CRV	20829

Reaction No. 74581



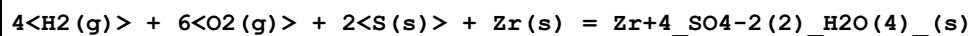
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:364.5 (4SF)	1	ELC	
2	25	0	Inf. Dilution	dG:-2082.4 (9.3MD) kJ	0	CRV	20829
3	25	0	Inf. Dilution	dH:-2359.8 (5.2MD) kJ	2	CRV	20829

Reaction No. 74582



No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:496.6 (4SF)	1	ELC	
2	25	0	Inf. Dilution	dG:-2842.1 (9.7MD) kJ	0	CRV	20829

Reaction No. 74583



No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	dG:-3008.8 (2.6MD) kJ	6	CRV	20829
2	25	0	Inf. Dilution	dH:-3470.9 (2.6MD) kJ	6	CRV	20829
3	25	0	Inf. Dilution	Cp:321.300 (0.3MD) J/K	6	CRV	20829

Reaction No. 74584

3<O2(g)> + 2<Se(s)> + Zr(s) = Zr(SeO3)2(s)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	dG:-1300.(4SF)kJ	1	EES	
2	25	0	Inf. Dilution	dH:-1609.6(2.7MD)kJ	6	CRV	20829

Reaction No. 74585							
1.843<Te(s)> + Zr(s) = ZrTe1.843(s)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	dG:-291.330(7.0MD)kJ	6	CRV	20829
2	25	0	Inf. Dilution	dH:-295.000(7.0MD)kJ	6	CRV	20829
3	25	0	Inf. Dilution	Cp:70.590(0.3MD)J/K	6	CRV	20829

Reaction No. 74586							
2<Te(s)> + Zr(s) = ZrTe2(s)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	dG:-290.040(7.0MD)kJ	6	CRV	20829
2	25	0	Inf. Dilution	dH:-294.000(7.0MD)kJ	6	CRV	20829

Reaction No. 74587							
4<Te(s)> + 5<Zr(s)> = Zr5Te4(s)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	dG:-810.430(44.0MD)kJ	6	CRV	20829
2	25	0	Inf. Dilution	dH:-829.000(44.0MD)kJ	6	CRV	20829
3	25	0	Inf. Dilution	Cp:222.000(0.3MD)J/K	6	CRV	20829

Reaction No. 74588							
4<O2(g)> + 3<Te(s)> + Zr(s) = ZrTe3O8(s)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	dG:-1700.(4SF)kJ	1	EES	
2	25	0	Inf. Dilution	dH:-2119.2(9.6MD)kJ	6	CRV	20829

Reaction No. 74589							
0.5<N2(g)> + Zr(s) = ZrN(s)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	dG:-341.850(1.9MD)kJ	6	CRV	20829
2	25	0	Inf. Dilution	dH:-370.500(1.4MD)kJ	6	CRV	20829
3	25	0	Inf. Dilution	Cp:39.700(1.4MD)J/K	6	CRV	20829

Reaction No. 74590							
$0.5\text{<N2(g)>} + 1.5\text{<O2(g)>} + \text{Zr(s)} - 3\text{<e-1>} = \text{Zr+4_NO3-1}$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:112.6(4SF)	1	ELC	
2	25	0	Inf. Dilution	dG:-648.380(9.2MD)kJ	0	CRV	20829

Reaction No. 74591							
$\text{N2(g)} + 3\text{<O2(g)>} + \text{Zr(s)} - 2\text{<e-1>} = \text{Zr+4_NO3-1(2)}$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	dG:-765.170(9.3MD)kJ	6	CRV	20829

Reaction No. 74592							
$3.5\text{<O2(g)>} + 2\text{<P(s)>} + \text{Zr(s)} = \text{ZrP2O7(s)}$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	Cp:150.700(1.5MD)J/K	6	CRV	20829

Reaction No. 74593							
$2\text{<H2(g)>} + 4.5\text{<O2(g)>} + 2\text{<P(s)>} + \text{Zr(s)} = \text{Zr(HPO4)2.H2O(s)}$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	dG:-3194.5(20.2MD)kJ	6	CRV	20829
2	25	0	Inf. Dilution	dH:-3466.1(1.6MD)kJ	6	CRV	20829

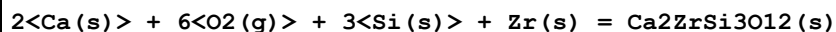
Reaction No. 74594							
$3\text{<H2(g)>} + 5\text{<O2(g)>} + 2\text{<P(s)>} + \text{Zr(s)} = \text{Zr(HPO4)2.2H2O(s)}$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	dG:-3000.(4SF)kJ	1	EES	
2	25	0	Inf. Dilution	dH:-3742.1(13.3MD)kJ	6	CRV	20829

Reaction No. 74595							
$2\text{<As(s)>} + 2\text{<H2(g)>} + 4.5\text{<O2(g)>} + \text{Zr(s)} = \text{Zr(HAsO4)2.H2O(s)}$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	dG:-2100.(4SF)kJ	1	EES	
2	25	0	Inf. Dilution	dH:-2659.3(12.0MD)kJ	6	CRV	20829

Reaction No. 74596							
$4\text{<C(s)>} + 6\text{<O2(g)>} + \text{Zr(s)} + 4\text{<e-1>} = \text{Zr+4_CO3-2(4)}$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #

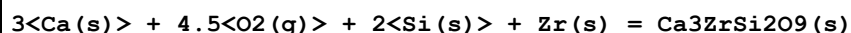
1	25	0	Inf. Dilution	dG:-2885.0 (10.9MD) kJ	6	CRV	20829
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Reaction No. 74597



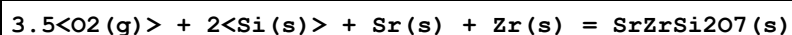
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	dG:-5951.5 (15.0MD) kJ	6	CRV	20829
2	25	0	Inf. Dilution	dH:-6283.0 (15.0MD) kJ	6	CRV	20829
3	25	0	Inf. Dilution	Cp:252.500 (2.5MD) J/K	6	CRV	20829

Reaction No. 74598



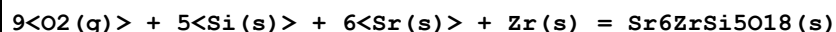
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	dG:-4769.3 (10.0MD) kJ	6	CRV	20829
2	25	0	Inf. Dilution	dH:-5029.0 (10.0MD) kJ	6	CRV	20829
3	25	0	Inf. Dilution	Cp:238.100 (1.5MD) J/K	6	CRV	20829

Reaction No. 74599



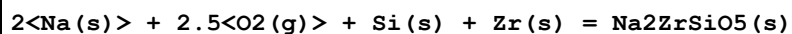
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	dG:-3444.0 (4.2MD) kJ	6	CRV	20829
2	25	0	Inf. Dilution	dH:-3640.8 (4.2MD) kJ	6	CRV	20829
3	25	0	Inf. Dilution	Cp:187.810 (0.4MD) J/K	6	CRV	20829

Reaction No. 74600



No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	dG:-7800. (4SF) kJ	1	EES	
2	25	0	Inf. Dilution	dH:-9867.9 (8.6MD) kJ	6	CRV	20829

Reaction No. 74601



No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	dG:-2523.5 (20.0MD) kJ	6	CRV	20829
2	25	0	Inf. Dilution	dH:-2670.0 (20.0MD) kJ	6	CRV	20829
3	25	0	Inf. Dilution	Cp:172.400 (2.0MD) J/K	6	CRV	20829

Reaction No. 74602

2<Na(s)> + 3.5<O2(g)> + 2<Si(s)> + Zr(s) = Na2ZrSi2O7(s)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	dG:-3412.4(10.0MD)kJ	6	CRV	20829
2	25	0	Inf. Dilution	dH:-3606.0(10.0MD)kJ	6	CRV	20829
3	25	0	Inf. Dilution	Cp:211.000(5.0MD)J/K	6	CRV	20829

Reaction No. 74603							
2<H2(g)> + 2<Na(s)> + 5.5<O2(g)> + 3<Si(s)> + Zr(s) = Na2ZrSi3O9.2H2O(s)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	dG:-4654.0(20.0MD)kJ	6	CRV	20829

Reaction No. 74604							
4<O2(g)> + 2<S(s)> + Zr(s) = Zr+4_SO4-2(2)_ (s)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	dG:-2009.5(2.1MD)kJ	6	CRV	20829
2	25	0	Inf. Dilution	dH:-2245.2(2.1MD)kJ	6	CRV	20829
3	25	0	Inf. Dilution	Cp:185.100(10.0MD)J/K	6	CRV	20829

Reaction No. 74605							
2<Na(s)> + 5.5<O2(g)> + 4<Si(s)> + Zr(s) = Na2ZrSi4O11(s)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	dG:-5181.0(20.000MD)kJ	6	CRV	20829

Reaction No. 74606							
3<H2(g)> + 2<Na(s)> + 9<O2(g)> + 6<Si(s)> + Zr(s) = Na2ZrSi6O15.3H2O(s)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	dG:-7931.0(30.000MD)kJ	6	CRV	20829

Reaction No. 74607							
4<Na(s)> + 6<O2(g)> + 3<Si(s)> + 2<Zr(s)> = Na4Zr2Si3O12(s)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	dG:-5944.0(20.057MD)kJ	6	CRV	20829
2	25	0	Inf. Dilution	dH:-6258.0(20.0MD)kJ	6	CRV	20829
3	25	0	Inf. Dilution	Cp:365.000(5.0MD)J/K	6	CRV	20829

Reaction No. 74608							
Na(s) + 6<O2(g)> + 3<P(s)> + 2<Zr(s)> = NaZr2P3O12(s)							

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	dG:-4557.0(20.2MD)kJ	6	CRV	20829
2	25	0	Inf. Dilution	dH:-4889.0(20.0MD)kJ	6	CRV	20829
3	25	0	Inf. Dilution	Cp:310.000(10.0MD)J/K	6	CRV	20829

Reaction No. 74609							
$2\text{Cs(s)} + 3.5\text{O}_2\text{(g)} + 2\text{Si(s)} + \text{Zr(s)} = \text{Cs}_2\text{ZrSi}_2\text{O}_7\text{(s)}$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	dH:-3587.0(10.0MD)kJ	6	CRV	20829

Reaction No. 74610							
$1.5\text{S(s)} + \text{Zr(s)} = \text{ZrS}_{1.5}\text{(s)}$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	dG:-502.110(3.9MD)kJ	6	CRV	20829
2	25	0	Inf. Dilution	dH:-509.400(1.1MD)kJ	6	CRV	20829

Reaction No. 74611							
$\text{H}_2\text{(g)} + 4\text{O}_2\text{(g)} + 2\text{P(s)} + \text{Zr(s)} = \text{Zr(HPO}_4)_2\text{(a.,s)}$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	dG:-2987.0(23.0MD)kJ	6	CRV	20829
2	25	0	Inf. Dilution	dH:-3166.2(6.0MD)kJ	6	CRV	20829

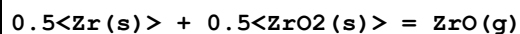
Reaction No. 74612							
$\text{H}_2\text{(g)} + 4\text{O}_2\text{(g)} + 2\text{P(s)} + \text{Zr(s)} = \text{Zr(HPO}_4)_2\text{(b.,s)}$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	dG:-2500.(4SF)kJ	1	EES	
2	25	0	Inf. Dilution	dH:-3159.1(6.0MD)kJ	6	CRV	20829

Reaction No. 74613							
$2\text{As(s)} + \text{H}_2\text{(g)} + 4\text{O}_2\text{(g)} + \text{Zr(s)} = \text{Zr(HAsO}_4)_2\text{(a.,s)}$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	dG:-1804.(4SF)kJ	1	EES	
2	25	0	Inf. Dilution	dH:-2364.7(12.1MD)kJ	6	CRV	20829

Reaction No. 74614							
$2\text{As(s)} + \text{H}_2\text{(g)} + 4\text{O}_2\text{(g)} + \text{Zr(s)} = \text{Zr(HAsO}_4)_2\text{(b.,s)}$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #

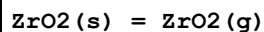
1	25	0	Inf. Dilution	dG:-1800.(4SF)kJ	1	EES	
2	25	0	Inf. Dilution	dH:-2360.9(12.0MD)kJ	6	CRV	20829

Reaction No. 74615



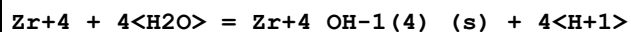
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	dG:550.(3SF)kJ	1	EES	
2	25	0	Inf. Dilution	dH:606.500(27.0MD)kJ	6	CRV	20829

Reaction No. 74616



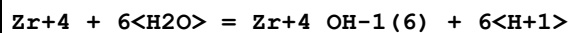
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:-130.3(4SF)	1	ELC	
2	25	0	Inf. Dilution	dH:811.000(47.0MD)kJ	4	CRV	20829

Reaction No. 74617



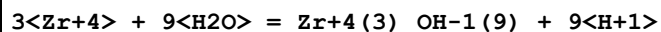
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:3.240(0.100MD)	6	CRV	20829
2	25	0	Inf. Dilution	lgK:2.77(0.13SD)	4	CRV	32222

Reaction No. 74618



No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:-33.2(3SF)	1	ELC	
2	25	0	Inf. Dilution	lgK:-29.000(0.700MD)	0	CRV	20829
3	25	0	Inf. Dilution	lgK:-31.96(0.35SD)	0	CRV	32222

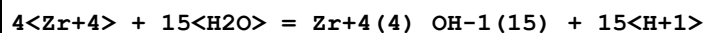
Reaction No. 74619



No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:12.190(0.080MD)	3	CRV	20829

Note (1): Found not to be required by [18RKA_31367]

Reaction No. 74620



No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
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1	25	0	Inf. Dilution	lgK:12.580(0.240MD)	3	CRV	20829
Note (1): Found not to be required by [18RKA_31367]							
2	25	0	Inf. Dilution	lgK:0.04(0.52SD)	0	CRV	32222

Reaction No. 74621							
$4\text{Zr}^{+4} + 16\text{H}_2\text{O} = \text{Zr}_4(\text{OH})_{16} + 16\text{H}^+$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:8.390(0.800MD)	3	CRV	20829
Note (1): Found not to be required by [18RKA_31367]							
2	25	0	Inf. Dilution	lgK:-4.36(0.61SD)	0	CRV	32222

Reaction No. 74622							
$\text{Zr}_4\text{F}_{14}(\text{s}) = \text{ZrF}_4(\text{g})$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	dH:240.180(0.040MD)kJ	3	CRV	20829

Reaction No. 74623							
$\text{Zr}_4\text{F}_{14}(\text{s}) = \text{ZrF}_4(\text{l})$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	dH:-64.220(0.420MD)kJ	3	CRV	20829

Reaction No. 74624							
$2\text{ZrCl}_3(\text{g}) = \text{ZrCl}_2(\text{g}) + \text{ZrCl}_4(\text{g})$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	dH:23.4000(7.900MD)kJ	6	CRV	20829

Reaction No. 74625							
$\text{Zr}_4\text{Cl}_{14}(\text{s}) = \text{ZrCl}_4(\text{g})$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:-9.3(2SF)	1	ELC	
2	25	0	Inf. Dilution	dH:109.300(0.040MD)kJ	6	CRV	20829

Reaction No. 74626							
$\text{Zr}_4\text{Br}_{14}(\text{s}) = \text{ZrBr}_4(\text{g})$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	dH:116.000(1.800MD)kJ	6	CRV	20829

Reaction No. 74627							
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Zr+4_I-1(4)_(s) = ZrI4(g)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:-13.122(0.064MD)	6	CRV	20829
2	25	0	Inf. Dilution	dH:133.400(0.300MD)kJ	6	CRV	20829

Reaction No. 74628							
0.5<N2(g)> + ZrO2(s) = O2(g) + ZrN(s)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	dH:730.100(1.200MD)kJ	6	CRV	20829

Reaction No. 74629							
H2O + 2<H+1(3)_PO4-3> + Zr+4 = 4<H+1> + Zr(HPO4)2.H2O(s)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:22.800(3.100MD)	6	CRV	20829

Reaction No. 74630							
Zr+4 + 4<CO3-2> = Zr+4_CO3-2(4)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:42.900(1.000MD)	6	CRV	20829
2	25	0	Inf. Dilution	lgK:42.4(0.1SD)	4	MSL	29848
3	25	0	Inf. Dilution	lgK:42.(2SF)	2	EOD	29848

Reaction No. 74631							
Zr+4 + H+1(4)_CDTA-4 = Zr+4_H+1(4)_CDTA-4							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	20	2	HClO4	lgK:29.88(4SF)	5	MIX	13428
2	25	0	Inf. Dilution	lgK:29.7(3SF)	1	ESC	

Reaction No. 74632							
Zr+4 + H+1_Glycolic-1 = Zr+4_H+1_Glycolic-1							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Unknown	lgK:7.7(2SF)	3	MKN	7753

Reaction No. 74633							
Zr+4 + H+1_PhenAcet-1 = Zr+4_H+1_PhenAcet-1							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Unknown	lgK:6.6(2SF)	3	MKN	7753

2	25	0.01	Unknown	lgK:6.8 (2SF)	3	MKN	7753
3	25	0.1	Unknown	lgK:6.2 (2SF)	3	MKN	7753

Reaction No. 74634							
Zr+4 + SCN-1 = Zr+4_SCN-1							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	23	0.1	HClO4	lgK:-2. (1SF)	0	MSP	13450
2	23	0.8	HClO4	lgK:-3.8 (2SF)	4	MSP	13450
3	25	0	Inf. Dilution	lgK:-2.6 (3SF)	2	EAE	

Reaction No. 74635							
Zr+4_SCN-1 + SCN-1 = Zr+4_SCN-1(2)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	23	0.1	HClO4	lgK:-1.4 (2SF)	0	MSP	13450
2	23	0.8	HClO4	lgK:-3.5 (2SF)	4	MSP	13450
3	25	0	Inf. Dilution	lgK:-2.6 (3SF)	2	EAE	

Reaction No. 74636							
Zr+4_SCN-1(2) + SCN-1 = Zr+4_SCN-1(3)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	23	0.1	HClO4	lgK:-1.25 (3SF)	0	MSP	13450
2	23	0.8	HClO4	lgK:-3.4 (2SF)	4	MSP	13450
3	25	0	Inf. Dilution	lgK:-2.8 (3SF)	2	EAE	

Reaction No. 74637							
Zr+4_SCN-1(3) + SCN-1 = Zr+4_SCN-1(4)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	23	0.1	HClO4	lgK:-1.1 (2SF)	0	MSP	13450
2	23	0.8	HClO4	lgK:-3.2 (2SF)	4	MSP	13450
3	25	0	Inf. Dilution	lgK:-2.9 (3SF)	2	EAE	

Reaction No. 74638							
Zr+4_SCN-1(4) + SCN-1 = Zr+4_SCN-1(5)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	23	0.1	HClO4	lgK:-1.1 (2SF)	0	MSP	13450
2	23	0.8	HClO4	lgK:-3.1 (2SF)	4	MSP	13450
3	25	0	Inf. Dilution	lgK:-3.1 (3SF)	2	EAE	

Reaction No. 74639							
Zr+4_SCN-1(5) + SCN-1 = Zr+4_SCN-1(6)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	23	0.1	HClO4	lgK:-1.0(2SF)	0	MSP	13450
2	23	0.8	HClO4	lgK:-2.9(2SF)	4	MSP	13450
3	25	0	Inf. Dilution	lgK:-3.2(3SF)	2	EAE	

Reaction No. 74640							
Zr+4_SCN-1(6) + SCN-1 = Zr+4_SCN-1(7)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	23	0.1	HClO4	lgK:-0.98(2SF)	0	MSP	13450
2	23	0.8	HClO4	lgK:-2.8(2SF)	4	MSP	13450
3	25	0	Inf. Dilution	lgK:-3.4(3SF)	2	EAE	

Reaction No. 74641							
Zr+4_SCN-1(7) + SCN-1 = Zr+4_SCN-1(8)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	23	0.1	HClO4	lgK:-0.96(2SF)	0	MSP	13450
2	23	0.8	HClO4	lgK:-2.7(2SF)	4	MSP	13450
3	25	0	Inf. Dilution	lgK:-3.61(3SF)	2	EAE	

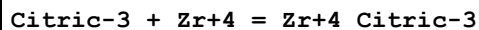
Reaction No. 74642							
Zr+4 + Malic-2 = Zr+4_Malic-2							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	23	2	HClO4	lgK:1.9(2SF)	3	MCE	7744
2	25	0	Inf. Dilution	lgK:3.84(3SF)	2	EAE	

Reaction No. 74643							
Zr+4 + Lactic-1 = Zr+4_Lactic-1							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	23	2	HClO4	lgK:2.27(3SF)	5	MCE	3262
2	23	2	HClO4	lgK:2.0(2SF)	3	MCE	7744
3	25	0	Inf. Dilution	lgK:3.17(3SF)	2	EAE	

Reaction No. 74644							
Zr+4 + Tartaric-2 = Zr+4_Tartaric-2							

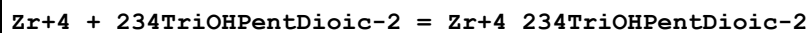
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	20	1	HCl	lgK:6.86(3SF)	3	MLC	4433
2	23	2	HClO4	lgK:2.2(2SF)	3	MCE	7744

Reaction No. 74645



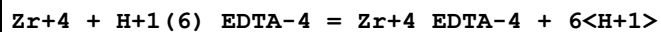
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	20	1	HCl	lgK:10.78(4SF)	3	MLC	4433
2	23	2	HClO4	lgK:3.1(2SF)	0	MCE	7744
3	25	0	Inf. Dilution	lgK:13.27(4SF)	4	CRV	31301

Reaction No. 74646



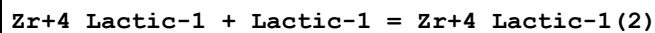
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	23	2	HClO4	lgK:3.41(3SF)	5	MCE	3262
2	23	2	HClO4	lgK:3.1(2SF)	3	MCE	7744

Reaction No. 74647



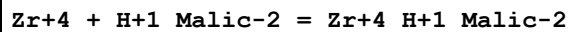
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	2	HClO4	dH:-5.86(0.15MD)	2	MCL	5106
2	25	3	HClO4	dH:-5.72(0.18MD)	2	MCL	5106
3	25	4	HClO4	dH:-5.66(0.16MD)	2	MCL	5106

Reaction No. 74648



No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	23	2	HClO4	lgK:2.54(3SF)	5	MCE	3262
2	25	0	Inf. Dilution	lgK:3.27(3SF)	2	EAE	

Reaction No. 74649



No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	23	2	HClO4	lgK:2.23(3SF)	5	MCE	3262
2	25	0	Inf. Dilution	lgK:3.2(3SF)	2	EAE	

Reaction No. 74650

Zr+4_234TriOHPentDioic-2 + 234TriOHPentDioic-2 = Zr+4_234TriOHPentDioic-2(2)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	23	2	HClO4	lgK:5.40(3SF)	5	MCE	3262

Reaction No. 74651							
Zr+4 + Glu-2 = Zr+4_Glu-2							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0.1	NaClO4	lgK:9.60(3SF)	4	MGL	22615

Reaction No. 74652							
Zr+4_Glu-2 + Glu-2 = Zr+4_Glu-2(2)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0.1	NaClO4	lgK:6.40(3SF)	4	MGL	22615

Reaction No. 74653							
Zr+4_Glu-2(2) + Glu-2 = Zr+4_Glu-2(3)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0.1	NaClO4	lgK:3.32(3SF)	4	MGL	22615

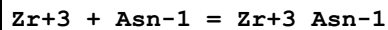
Reaction No. 74654							
Zr+4 + H+1(4)_EDTA-4 = Zr+4_EDTA-4 + 4<H+1>							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:10.3(3SF)	1	ELC	
2	25	0.01	HClO4	lgR:19.4(3SF)	3	MMX	13427

Reaction No. 74855							
Zr+4 + e-1 = Zr+3							
Note: Zirconium exists in aqueous solutions in the + 4 oxidation state only. See Brown et al., OECD Chem. Thermodynamics Ser., Vol. 8, 2005, p. 97							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:-96.0(3SF)	0	EOD	22560
Note (1): From Latimer WM, The Oxidation States of the Elements, 1952							
2	25	0	Inf. Dilution	E:-1.430(4SF)	0	MEF	22560
Note (2): From Latimer WM, The Oxidation States of the Elements, 1952							

Reaction No. 75034							
CrO4-2 + Zr+4 = Zr+4_CrO4-2							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #

1	0.01	0	Inf. Dilution	lgK:7.0102(5SF)	3	EOD	22958
2	25	0	Inf. Dilution	lgK:7.3125(5SF)	3	EOD	22958
3	60	0	Inf. Dilution	lgK:7.9602(5SF)	3	EOD	22958
4	100	0	Inf. Dilution	lgK:8.8346(5SF)	3	EOD	22958
5	150	0	Inf. Dilution	lgK:10.048(5SF)	3	EOD	22958
6	200	0	Inf. Dilution	lgK:11.3930(6SF)	3	EOD	22958

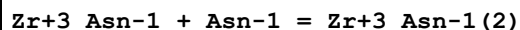
Reaction No. 75792



Note: Zirconium exists in aqueous solutions in the + 4 oxidation state only. See Brown et al., OECD Chem. Thermodynamics Ser., Vol. 8, 2005, p. 97

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0.1	NaClO4	lgK:8.80(3SF)	0	MGL	22630

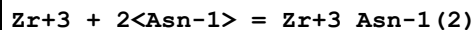
Reaction No. 75793



Note: Zirconium exists in aqueous solutions in the + 4 oxidation state only. See Brown et al., OECD Chem. Thermodynamics Ser., Vol. 8, 2005, p. 97

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0.1	NaClO4	lgK:6.25(3SF)	0	MGL	22630

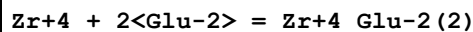
Reaction No. 75794



Note: Zirconium exists in aqueous solutions in the + 4 oxidation state only. See Brown et al., OECD Chem. Thermodynamics Ser., Vol. 8, 2005, p. 97

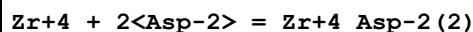
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0.1	NaClO4	lgK:15.05(4SF)	0	MGL	22630

Reaction No. 75821



No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0.1	NaClO4	lgK:16.00(4SF)	4	MGL	22615

Reaction No. 75847



No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0.1	NaClO4	lgK:16.55(4SF)	4	MGL	22615

Reaction No. 75856							
$\text{Zr}+4 + 2\langle\text{Gln-1}\rangle = \text{Zr}+4_ \text{Gln-1}(2)$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0.1	NaClO4	lgK:14.85 (4SF)	4	MGL	22630

Reaction No. 76500							
$\text{Zr}+4 + 6\langle\text{OH-1}\rangle = \text{Zr}+4_ \text{OH-1}(6)$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:55. (2SF)	0	ELC	
2	25	0	Inf. Dilution	lgK:48. (2SF)	0	EOD	29848
3	25	0	Inf. Dilution	lgK:50.72 (0.24SD)	4	CRV	31367
Note (3): SIT fit							
4	25	0	Inf. Dilution	lgK:50.95 (0.35SD)	4	CRV	31367
Note (4): Pitzer fit							

Reaction No. 78022							
$2\langle\text{EDTA-4}\rangle + 2\langle\text{Zr}+4\rangle = \text{Zr}+4(2)_ \text{EDTA-4}(2)$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:70.0 (4SF)	1	ELC	

Reaction No. 78032							
$\text{Citric-3} + \text{Zr}+4 + \text{H}+1 = \text{Zr}+4_ \text{H}+1_ \text{Citric-3}$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:14.9 (3SF)	1	ELC	

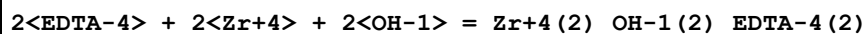
Reaction No. 78042							
$\text{Citric-3} + \text{Zr}+4 + 2\langle\text{H}+1\rangle = \text{Zr}+4_ \text{H}+1(2)_ \text{Citric-3}$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:18.27 (4SF)	1	ELC	

Reaction No. 78045							
$\text{Gly-1} + \text{Zr}+4 + \text{H}+1 = \text{Zr}+4_ \text{H}+1_ \text{Gly-1}$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:13.8 (4SF)	1	ELC	

Reaction No. 78114							
$\text{EDTA-4} + \text{Zr}+4 + \text{OH-1} = \text{Zr}+4_ \text{OH-1}_ \text{EDTA-4}$							

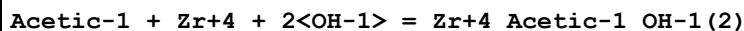
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1	25	0	Inf. Dilution	lgK:40.84 (4SF)	1	ELC	

Reaction No. 78164



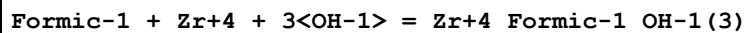
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:84.97 (4SF)	1	ELC	

Reaction No. 78196



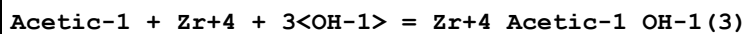
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:36.13 (4SF)	1	ELC	

Reaction No. 78198



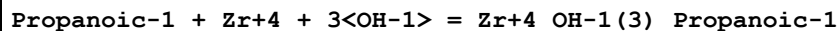
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:44.73 (4SF)	1	ELC	

Reaction No. 78212



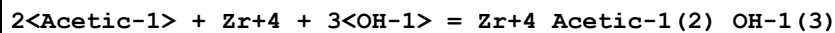
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:44.68 (4SF)	1	ELC	

Reaction No. 78225



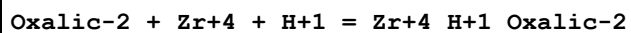
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:45.13 (4SF)	1	ELC	

Reaction No. 78226



No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:46.51 (4SF)	1	ELC	

Reaction No. 78277



No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:10.84 (4SF)	1	ELC	

Reaction No. 78328							
$2\text{<Oxalic-2>} + \text{Zr+4} + 2\text{<H+1>} = \text{Zr+4_H+1(2)_Oxalic-2(2)}$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:20.76(4SF)	1	ELC	

Reaction No. 78412							
$4\text{<Oxalic-2>} + \text{Zr+4} + 4\text{<H+1>} = \text{Zr+4_H+1(4)_Oxalic-2(4)}$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:28.31(4SF)	1	ELC	

Reaction No. 78626							
$\text{Ala-1} + \text{Zr+4} + \text{H+1} = \text{Zr+4_H+1_Ala-1}$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:14.04(4SF)	1	ELC	

Reaction No. 78632							
$\text{EDTA-4} + \text{Zr+4} + \text{H+1} = \text{Zr+4_H+1_EDTA-4}$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:39.35(4SF)	1	ELC	

Reaction No. 80690							
$\text{ZrO2(am.,s)} + 2\text{<H2O>} = \text{Zr+4} + 4\text{<OH-1>}$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:-56.19(0.03SD)	3	CRV	31367

Reaction No. 80691							
$\text{Zr+4} + 6\text{<OH-1>} + 3\text{<Ca+2>} = \text{Zr+4_Ca+2(3)_OH-1(6)}$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:57.00(0.22SD)	4	CRV	31367
Note (1): SIT fit							
2	25	0	Inf. Dilution	lgK:57.47(0.20SD)	4	CRV	31367
Note (2): Pitzer fit							

Reaction No. 81029							
$\text{Zr+4} + 4\text{<H2O>} = \text{Zr+4_OH-1(4)_(am.,s)} + 4\text{<H+1>}$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:0.14(0.13SD)	4	CRV	32222

Reaction No. 81334							
$\text{BaZrO}_3(\text{s}) + 6\text{H}^+ = \text{Ba}^{2+} + \text{Zr}^{4+} + 3\text{H}_2\text{O}$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	$\lg K: 27.45 (4\text{SF})$	4	CRV	32224

Reaction No. 81343							
$\text{CaO} \cdot \text{ZrO}_2(\text{s}) + 6\text{H}^+ = \text{Ca}^{2+} + \text{Zr}^{4+} + 3\text{H}_2\text{O}$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	$\lg K: 28.54 (4\text{SF})$	4	CRV	32224

Reaction No. 81614							
$\text{SrZrO}_3(\text{s}) + 6\text{H}^+ = \text{Sr}^{2+} + \text{Zr}^{4+} + 3\text{H}_2\text{O}$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	$\lg K: 20.0 (3\text{SF})$	1	ELC	
2	25	0	Inf. Dilution	$\lg K: 29.28 (4\text{SF})$	0	CRV	32224

Reaction No. 81707							
$\text{ZrH}_2(\text{s}) = \text{Zr}^{4+} + 2\text{H}^+ + 6\text{e}^-$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	$\lg K: 70.3 (3\text{SF})$	1	ELC	
2	25	0	Inf. Dilution	$\lg K: 78.88 (4\text{SF})$	0	CRV	32224

Reaction No. 81708							
$3\text{Zr}^{4+} + 5\text{H}_2\text{O} = \text{Zr}_3(\text{OH})_5 + 5\text{H}^+$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	$\lg K: 3.7 (2\text{SF})$	2	CRV	32224

Reaction No. 81709							
$\text{ZrSiO}_4(\text{s}) + 4\text{H}^+ = \text{Zr}^{4+} + \text{H}_2\text{SiO}_4^{2-}$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	$\lg K: -14.8 (3\text{SF})$	1	ELC	
2	25	0	Inf. Dilution	$\lg K: -5.64 (3\text{SF})$	0	CRV	32224

Data Assessment

Weight	Assessment
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9	highest accuracy
8	multi-determined; excellent dependability
7	trustworthy; outstanding single determination
6	better than average reliability
5	average single determination
4	some doubt
3	poorly known
2	guess to achieve consistency (or just as bad)
1	no information but consistent
0	problematic/inconsistent

Techniques

Abbrev	Method
CMN	Several experimental values (crit.)
CRV	Critical review
CSM	Smith & Martell Vols 1-6 (crit.)
EAE	Automated extrapolation
EDH	Debye-Hueckel corrections
EES	Personal estimate
EFE	Free Energy relations
EIU	Critical IUPAC review (estim.)
ELC	Linear Combination of Reactions
ENS	Not specified
EOD	Calculated from data of others
EOR	Other Reaction Constants
ERS	Rough subjective
ESC	Standard conditions anchor
MCE	Cation exchange
MCG	Chromatography, Gel
MCL	Calorimetry
MDG	Combination of Thermodynamic Data
MDS	Distribution between two phases
MEF	Electromotoric force, not specified

MGL	Glass electrode
MHY	Emf with hydrogen electrode
MIS	Ion selective electrode
MIX	Ion exchange
MKN	Rate of reaction
MLC	Ligand competition
MMX	Mixed techniques
MPL	Polarography
MPT	Potentiometry
MQH	Quinhydrone electrode
MRX	Emf with redox electrode
MSC	Miscellaneous
MSE	Solvent Extraction
MSL	Solubility
MSP	Spectroscopy
MTD	Temperature difference

References

Reference No.: 8(76SmM)

Critical Stability Constants, Vol. 4, 1976 Smith RM; Martell AE; Plenum, New York

Reference No.: 138(67Tik)

Russ. J. Inorg. Chem., 1967, 12, 494 - 497; Tikhonova LI; Complex Formation by Zirconium with Polyaminopolyacetic Acids

Reference No.: 140(64SiM)

Stability Constants of Metal-ion Complexes SP17, 1964 Sillen LG; Martell AE; The Chemical Soc., London

Reference No.: 142(67BoA)

Helv. Chim. Acta, 1967, 50, 2349 - 2356; Bottari E; Anderegg G; Complexons. XLII. Study of 1:1 Complexes of some Tri- and Tetravalent Metal Ions with Polyaminecarboxylates by Redox Measurement

Reference No.: 145(69BNS)

J. Am. Chem. Soc., 1969, 91, 4680 - 4683; Brunetti AP; Nancollas GH; Smith PN; Thermodynamics of Ion Association. XIX. Complexes of Divalent Metal Ions with Monoprotonated Ethylenediaminetetraacetate

Reference No.: 437(80BoH)

IUPAC Chemical Data Series No. 27, 1980 Bond AM; Hefter GT; Critical Survey of

Stability Constants and Related Thermodynamic Data of Fluoride Complexes in Aqueous Solution; Pergamon, Oxford, U.K.

Reference No.: 552(95SMM)

NIST Critical Stability Constants of Metal Complexes Database, 1995 Smith RM; Martell AE; Motekaitis RJ; Version 2.0; National Institute of Standards and Technology, Gaithersburg, USA

Reference No.: 773(85KoS)

Handbook of Chemical Equilibria in Analytical Chemistry, 1985 Kotrly S; Sucha L; Ellis Horwood Series in Anal. Chem.; Eds. Chalmers RA & Masson M; Ellis Horwood, Chichester

Reference No.: 802(90Wea)

CRC Handbook of Chemistry and Physics, 70th Edn., 1990 Weast RC; A Ready-Reference Book of Chemical and Physical Data; CRC Press, Cleveland, Ohio

Reference No.: 903(77And)

Critical Survey of Stability Constants of EDTA Complexes, 1977 Anderegg G; IUPAC Chemical Data Series. 14; Pergamon Press, Oxford

Reference No.: 905(95Ber)

Pure Appl. Chem., 1995, 67, 1117 - 143; Berthon G; Critical Review of Formation Constants of Selected Amino Acids with Polar Side Chains

Reference No.: 906(79Per)

IUPAC Chemical Data Series No. 22, 1979 Perrin DD; Stability Constants of Metal-Ion Complexes, First Edition Part B, Organic Ligands; Pergamon, Oxford

Reference No.: 931(71SiM)

Stability Constants of Metal-ion Complexes SP25, 1971 Sillen LG; Martell AE; Supplement No 1 to Special Publication No 25; Chemical Society, London

Reference No.: 999(91MaM)

Estimation of Constants for JESS Database, 1991 May PM; May RF

Reference No.: 1118(82And)

Pure Appl. Chem., 1982, 54, 2693 - 2758; Anderegg G; Critical Survey of Stability Constants of NTA Complexes

Reference No.: 1539(78Kra)

Atlas of Metal-Ligand Equilibria in Aqueous Solution, 1978 Kragten J; Ellis Horwood, Chichester UK

Reference No.: 1555(91KSG)

Pure Appl. Chem., 1991, 63, 597 - 638; Kiss T; Sovago I; Gergely A; Critical Survey of Stability Constants of Complexes of Glycine

Reference No.: 1562(76AnM)

Helv. Chim. Acta, 1976, 59, 1498 - 1511; Anderegg G; Malik SC; Komplexe XLVII. The Stability of Palladium(II) Complexes with Aminopolycarboxylate Anions

Reference No.: 2261(93PeP)

IUPAC Stability Constants Database, 1993 Pettit LD; Powell HKJ; IUPAC (in co-operation with Academic Software), Otley, UK

Reference No.: 2373(89And)

Comprehensive Coordination Chemistry, Vol. 2, 1989, 777 - 792; Anderegg G; Complexones

Reference No.: 2392(93SKG)

Pure Appl. Chem., 1993, 65, 1029 - 1080; Sovago I; Kiss T; Gergely A; Critical Survey of the Stability Constants of Complexes of Aliphatic Amino Acids

Reference No.: 2556(93MSM)

NIST Critical Stability Constants of Metal Complexes Database, 1993 Martell AE; Smith RM; Motekaitis RJ; Version 1.0; National Institute of Standards and Technology, Gaithersburg, USA

Reference No.: 3262(64RME)

J. Inorg. Nucl. Chem., 1964, 26, 965 - 980; Ryabchikov DI; Marov IN; Ermakov AN; Belyaeva VK; Stability of Some Inorganic and Organic Complex Compounds of Zirconium and Hafnium

Reference No.: 4210(65BaG)

J. Inorg. Nucl. Chem., 1965, 27, 1085 - 1092; Baroncelli F; Grossi G; The Complexing Power of Hydroxamic Acids and its Effect on the Behaviour of Organic Extractants in the Reprocessing of Irradiated Fuels - I

Reference No.: 4405(66EME)

Russ. J. Inorg. Chem., 1966, 11, 618 - 620; Ermakov AN; Marov IN; Evtikova GA; Zirconium and Hafnium Complexonates

Reference No.: 4406(66PRS)

Russ. J. Inorg. Chem., 1966, 11, 621 - 624; Pakhomova AD; Rudyakova YaL; Sheka IA; Spectrophotometry of Complex Formation of Zirconium and Hafnium Ions with Pyrogallol

Reference No.: 4433(66KSG)

Russ. J. Inorg. Chem., 1966, 11, 1485 - 1488; Korenman IM; Sheyanova FR; Gur'eva ZM; Instability Constants of Complexes of Zirconium with Some Organic Ligands

Reference No.: 4444(66SoI)

Russ. J. Inorg. Chem., 1966, 11, 1013 - 1016; Solovkin AS; Ivantsov AI; Hydrolysis Constants of the Zr+4 Ion in Perchlorate Media

Reference No.: 4695(75BuS)

J. Chem. Soc., Perkin Trans. 2, 1975,, 1110 - 1112; Bukka K; Satchell RS; Quantitative Aspects of Lewis Acidity. Part XIV. Comparison of the Acidity of Tin Tetrachloride, Tellurium Tetrachloride, and Zirconium Tetrabromide towards Substituted Anilines in Dioxan

Reference No.: 4708(76TrS)

J. Inorg. Nucl. Chem., 1976, 38, 145 - 148; Tribalat S; Schriver L; Etude de Quelques Complexes du Zirconium(IV) et du Hafnium(IV) au Moyen de L'Extraction des Isopropyltropolonates par le Chloroforme

Reference No.: 4923(90AHN)

Acta Chem. Scand., 1990, 44, 1 - 7; Ahrlund S; Hefter G; Noren B; A Calorimetric Study of the Mononuclear Fluoride Complexes of Zirconium(IV), Hafnium(IV), Thorium(IV) and Uranium(IV)

Reference No.: 5076(61PMZ)

Russ. J. Inorg. Chem., 1961, 6, 630 - 634; Peshkova VM; Mel'chakova NV; Zhemchuzhin SG; Complex Formation in the Benzoylacetone[1-phenylbutane-1,3-Dione] Zirconium-benzene-water System and the Hydrolysis of Zirconium Ions

Reference No.: 5077(73Nor)

Acta Chem. Scand., 1973, 27, 1369 - 1384; Noren B; The Hydrolysis of Zr+4 and Hf+4

Reference No.: 5106(78VLL)

Russ. J. Inorg. Chem., 1978, 23, 29 - 32; Vasil'ev VP; Lymar VP; Lytkin AI;
Thermodynamic Characteristics of the Reaction of Zirconium(IV) with
Ethylenediaminetetra-acetic Acid

Reference No.: 5398(82Hog)

IUPAC Chemical Data Series No. 21, 1982 Hogfeldt E; Stability Constants of Metal-ion
Complexes, 2nd Suppl., Part A, Inorganic Ligands; Pergamon, Oxford

Reference No.: 5458(80BeT)

Report LBL-11448, 1980 Benson LV; Teague LS; Tabulation of Thermodynamic Data for
Chemical Reactions Involving 58 Elements Common to Radioactive Waste

Reference No.: 5502(57Sol)

Zh. Neorg. Khim., 1957, 2, 611 - 236; Solovkin AS; Determination of Hydrolysis
Constants and of Formation Constants of Zr+4 with Nitrate- and Chloride- Ions by
Extraction Methods

Reference No.: 5987(97LPP)

Pure Appl. Chem., 1997, 69, 329 - 381; Lajunen LHJ; Portanova R; Piispanen J; Tolazzi
M; Stability Constants for alpha-Hydroxycarboxylic Acid Complexes with Protons and
Metal Ions and the Accompanying Enthalpy Changes. Part I: Aromatic ortho-
Hydroxycarboxylic Acids

Reference No.: 6330(95LiF)

CRC Handbook of Chemistry and Physics, 76th Edn., 1995 Lide DR; Frederikse HPR; A
Ready-Reference Book of Chemical and Physical Data; CRC Press, Cleveland, Ohio, U.S.A.

Reference No.: 6422(95BaP)

Thermochemical Data of Pure Substances, 3rd Edn., 1995 Barin I; Platzki G; Weinheim,
Germany

Reference No.: 6478(81BaM)

Am. J. Sci., 1981, 281, 935 - 962; Baes CF Jr; Mesmer RE; The Thermodynamics of Cation
Hydrolysis

Reference No.: 6494(95RoH)

U.S. Geol. Surv. Bull. 2131, 1995 Robie RA; Hemingway BS; Thermodynamic Properties of
Minerals and Related Substances at 298.15 K and 1 Bar (10⁵ Pascals) Pressure and at
Higher Temperatures; U.S. Government Printing Office, Washington, U.S.A

Reference No.: 6496(96StM)

Aquatic Chemistry, 3rd Edn., 1996 Stumm W; Morgan JJ; Chemical Equilibria and Rates in
Natural Waters; John Wiley & Sons, Inc., New York

Reference No.: 7033(95AWW)

Appl. Geochem., 1995, 10, 603 - 620; Aja SU; Wood SA; Williams-Jones AE; The Aqueous
Geochemistry of Zr and the Solubility of some Zr-bearing Minerals

Reference No.: 7421(97Mar)

Ion Properties, 1997 Marcus Y; Marcel Dekker Inc., New York, U.S.A.

Reference No.: 7744(60REB)

Russ. J. Inorg. Chem., 1960, 5, 505 - 513; Ryabchikov DI; Ermakov AN; Belyaeva VK;
Marov IN; Zirconium and Hafnium Complexes with Some Hydroxy Acids

Reference No.: 7753(73KPK)

Russ. J. Inorg. Chem., 1973, 18, 776 - 779; Kalinina SS; Prozorovskaya ZN; Komissarova

LN; Yuranova LI; Spitsyn VI; Kinetic Study of Complex Formation by Zirconium and Hafnium with Glycollic and Phenylacetic Acids

Reference No.: 7761(89Bra)

J. Phys. Chem. Ref. Data, 1989, 18, 1 - 21; Bratsch SG; Standard Electrode Potentials and Temperature Coefficients in Water at 298.15 K

Reference No.: 7998(49CoM)

J. Am. Chem. Soc., 1949, 71, 3182 - 3191; Connick RE; McVey WH; The Aqueous Chemistry of Zirconium

Reference No.: 8136(53WhD)

J. Am. Chem. Soc., 1953, 75, 3081 - 3085; Whiteker RA; Davidson N; Ion-Exchange and Spectrophotometric Investigation of Iron(III) Sulfate Complex Ions

Reference No.: 9029(97SSW)

Geochim. Cosmochim. Acta, 1997, 61, 907 - 950; Shock EL; Sassani DC; Willis M; Sverjensky DA; Inorganic Species in Geologic Fluids: Correlations among Standard Molal Thermodynamic Properties of Aqueous Ions and Hydroxide Complexes

Reference No.: 11516(00PeP)

IUPAC Stability Constants Database, 2000 Pettit LD; Powell HKJ; Release 4; Academic Software, U.K.

Reference No.: 13242(05AAD)

Pure Appl. Chem., 2005, 77, 1445 - 1495; Anderegg G; Arnaud-Neu F; Delgado R; Felcman J; Popov K; Critical Evaluation of Stability Constants of Metal Complexes of Complexones for Biomedical and Environmental Applications

Reference No.: 13427(64InM)

Inorg. Chem., 1964, 1, 81 - 87; Intorre BJ; Martell AE; Zirconium Complexes in Aqueous Solution. III. Estimation of Formation Constants

Reference No.: 13428(70PrH)

J. Inorg. Nucl. Chem., 1970, 32, 953 - 960; Prasilova J; Havlicek J; Determination of Stability Constants of some Complexes of Zirconium using Dinonyl Naphthalene Sulphonic Acid as Liquid Ion Exchanger

Reference No.: 13450(66GoS)

Russ. J. Inorg. Chem., 1966, 11, 419 - 421; Golub AM; Sergun'kin VN; Formation of Thiocyanato-Complexes of Zirconium and Hafnium in Aqueous Solution

Reference No.: 13458(69Nor)

Acta Chem. Scand., 1969, 23, 379 - 387; Noren B; Solvent Extraction Studies of the Fluoride and Sulphate Complexes of Zirconium(IV)

Reference No.: 13652(59RyS)

Acta Chem. Scand., 1959, 13, 2057 - 2069; Rydberg J; Sullivan JC; Least Squares Computer Calculations of Equilibrium Constants from Solvent Extraction Data

Reference No.: 16981(11OrB)

Russ. J. Inorg. Chem., 2011, 56, 975 - 980; Orlov YF; Belkina EI; Correlations between the Stability Constants of Metal Hydroxo Complexes and the Solubility Products of Crystalline Hydroxides. Series of the Polarizing Effect of Metal Cations

Reference No.: 19258(62REB)

Russ. J. Inorg. Chem., 1962, 7, 34 - 37; Ryabchikov DI; Ermakov AN; Belyaeva VK; Marov IN; Yao K'o-min; Ion Exchange Study of Complex Formation by Zirconium and Hafnium with Sulphate Ions

Reference No.: 20289(93Arc)

J. Phys. Chem. Ref. Data, 1993, 22, 1441 - 1453; Archer DG; Thermodynamic Properties of Synthetic Sapphire (α -Al₂O₃), Standard Reference Material 720 and the Effect of Temperature-scale Differences on Thermodynamic Properties

Reference No.: 20829(05BCG)

OECD Chemical Thermodynamics Series, Volume 8, 2005 Brown PL; Curti E; Grambow B; Ekberg C; Chemical Thermodynamics of Zirconium; Elsevier Science

Reference No.: 22519(05Lid)

CRC Handbook of Chemistry and Physics, 86th Edn., 2005 Lide DR; A Ready-Reference Book of Chemical and Physical Data; Taylor & Francis, Boca Raton, USA

Reference No.: 22551(06Inz)

Encyclopedia of Electrochemistry, Vol. 7a, 2006, 17 - 75; Inzelt G; Standard, Formal and Other Characteristic Potentials of Selected Electrode Reactions

Reference No.: 22560(05PeP)

IUPAC Stability Constants Database, 2005 Pettit LD; Powell HKJ; Release 4.71; Academic Software, U.K.

Reference No.: 22615(72SiS)

Talanta, 1972, 19, 699 - 700; Singh MK; Srivastava MN; Potentiometric Determination of Stepwise Stability Constants of Zirconium and Thorium Chelates Formed with Aspartic and Glutamic Acids

Reference No.: 22616(63AKN)

Acta Chem. Scand., 1963, 17, 411 - 424; Ahrlund S; Karipides D; Noren B; The Fluoride and Sulphate Complexes of Zirconium(IV)

Reference No.: 22619(62MaR)

Russ. J. Inorg. Chem., 1962, 7, 533 - 539; Marov IN; Ryabchikov DI; Complex Formation of Zr(IV) and Hf(IV) with Chloride, Nitrate, and Oxalate Ions

Reference No.: 22620(64EME)

Russ. J. Inorg. Chem., 1964, 9, 275 - 279; Ermakov AN; Marov IN; Evtikova GA; Complex Formation by Zirconium and Hafnium with Nitrilotriacetic Acid

Reference No.: 22630(73TeS)

Talanta, 1973, 20, 360 - 361; Tewari RC; Srivastava MN; Potentiometric Determination of Stepwise Stability Constants of Zirconium, Thorium and Uranium Chelates of Asparagine and Glutamine

Reference No.: 22958(10AML)

Appl. Geochem., 2010, 25, 242 - 260; Accornero M; Marini L; Lelli M; Prediction of the Thermodynamic Properties of Metal-chromate Aqueous Complexes to High Temperatures and Pressures and Implications for the Speciation of Hexavalent Chromium in Some Natural Waters

Reference No.: 23624(97Ric)

The Chemistry of Aqua Ions, 1997 Richens DT; Synthesis, Structure and Reactivity. A Tour Through the Periodic Table of the Elements; John Wiley & Sons, Chichester, U.K.

Reference No.: 29487(09CDF)

Geochim. Cosmochim. Acta, 2009, 73, 2283 - 2298; Curti E; Dahn R; Farges F; Vespa M; Na, Mg, Ni and Cs Distribution and Speciation after Long-term Alteration of a Simulated Nuclear Waste Glass: A Micro-XAS/XRF/XRD and Wet Chemical Study

Reference No.: 29848(17KoS)

J. Solution Chem., 2017, 46, 1741 - 1759; Kobayashi T; Sasaki T; Solubility of Zr(OH)₄(am) and the Formation of Zr(IV) Carbonate Complexes in Carbonate Solutions Containing 0.1-5.0 mol.dm⁻³ NaNO₃

Reference No.: 29883(17MRD)

Coord. Chem. Rev., 2017, 352, 499 - 516; McInnes LE; Rudd SE; Donnelly PS; Copper, Gallium and Zirconium Positron Emission Tomography Imaging Agents: The Importance of Metal Ion Speciation

Reference No.: 31291(84PSG)

U.S. Bur. Mines Bull. 677, 1984 Pankratz LB; Stuve JM; Gokcen NA; Thermodynamic Data for Mineral Technology; US Det. Interior, Bur. Mines, Washington, DC, USA

Reference No.: 31301(10KFD)

Japan Atomic Energy Agency Data/Code 2009-024, 2010 Kitamura A; Fujiwara K; Doi R; Yoshida Y; Mihara M; Terashima M; Yui M; JAEA Thermodynamic Database for Performance Assessment of Geological Disposal of High-level Radioactive and TRU Wastes; Geological Isolation Research Unit, JAEA, Japan

Reference No.: 31367(18RKA)

J. Solution Chem., 2018, 47, 855 - 891; Rai D; Kitamura A; Altmaier M; Rosso KM; Sasaki T; Kobayashi T; A Thermodynamic Model for ZrO₂(am) Solubility at 25 C in the Ca²⁺ - Na⁺ - H⁺ - Cl⁻ - OH⁻ - H₂O System: A Critical Review

Reference No.: 31756(19LFZ)

Dalton Trans., 2019, 48, 17898 - 17907; Lehmann S; Foerstendorf H; Zimmermann Th; Patzschke M; Bok F; Brendler V; Stumpf T; Steudtner R; Thermodynamic and Structural Aspects of the Aqueous Uranium(IV) System - Hydrolysis vs. Sulfate Complexation

Reference No.: 32222(06DGC)

Technical Report TR-06-17, 2006 Duro L; Grive M; Cera E; Domenech C; Bruno J; Update of a Thermodynamic Database for Radionuclides to Assist Solubility Limits Calculation for Performance Assessment; Swedish Nucl. Fuel & Waste Management Co, Stockholm, Sweden

Reference No.: 32224(80BeT)

Lawrence Berkeley Laboratory Report LBL-11448, 1980 Benson LV; Teague LS; A Tabulation of Thermodynamic Data for Chemical Reactions Involving 8 Elements Common to Radioactive Waste Package Systems